OLS → ordinary least square

Date | Page |

① **OLS** linear regression Hypothesis & Squared Error cost :
state hypothesis for m samples & d feature (with intercept)..

concept → $Y = X\beta + \varepsilon$

ε → noise , β → Parameters

$E[\varepsilon] = 0$, $Cov(\varepsilon) = E(\varepsilon\varepsilon^T) = \sigma^2 I$

Assumption : since our errors for each samples are independ from another .

$\varepsilon\varepsilon^T = diag(\varepsilon_1^2, \varepsilon_2^2 \cdots, \varepsilon_m^2)$

$\varepsilon_i^2 = \sigma^2 = diag(\sigma^2, \sigma^2 \cdots, \sigma^2)$

$= \sigma^2 \cdot I$

$E[constant] = constant$

$Y \in R^{m\times 1}$

$X \to [1\ X_1\ X_2 \cdots X_n]$

$I \to$ col^m of 1 → intercept

$X \in R^{m\times d}$ $n+1 = d$

$\beta \in R^{d\times 1}$, $\varepsilon \in R^{m\times 1}$

do, $h_\beta(x) = X\beta$ → output for input X (prediction)

We know the ultimately cost fun^n can be written as

→ $J(\beta)$ = square error average

$= \frac{1}{2m} \sum_{i=1}^{m} (y^{(i)} - x^{(i)}\beta)^2$

$\beta = [\beta_0\ \beta_1 \cdots \beta_n]^T$

$(x^{(i)}, y^{(i)})_{i=1}^{m}$

$x^{(i)} \in R^{1\times d}$

$\beta \in R^{d\times 1}$

→ $J(\beta) = \frac{1}{2m} (Y - X\beta)^T (Y - X\beta)$

But why square term not anything else
lets look at this :

$J(\beta) = \frac{1}{m} \sum_{i=1}^{m} cost(h_\beta(x^{(i)}), y^{(i)})$

$cost(h_\beta(x^{(i)}), y^{(i)})$?

$Y = X\beta + \varepsilon$ $\quad \varepsilon_i \sim N(0, \sigma^2)$

$\varepsilon_i = Y - X\beta$ $\quad f_\varepsilon(\varepsilon_i) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{\varepsilon_i^2}{2\sigma^2}\right)$

$P(\varepsilon_i) = f_\varepsilon(\varepsilon_i)$ $\quad \varepsilon_i = Y^{(i)} - X\beta$

So, likelihood : $P(\vec{\varepsilon}; h_\beta(X), y) = L(h_\beta(X), y) = L(\beta)$

$L(\beta) = P(\varepsilon_1) P(\varepsilon_2) \cdots P(\varepsilon_m)$

$= \prod_{i=1}^{m} P(\varepsilon_i)$

$\log L(\beta) = l(\beta) = \sum_{i=1}^{m} \log(P(\varepsilon_i))$

$l(\beta) = \sum_{i=1}^{m} \left(\log\left(\frac{1}{\sqrt{2\pi}\sigma}\right) - \frac{(y^{(i)} - X^{(i)}\beta)^2}{2\sigma^2} \right)$

$l(\beta) = \underbrace{m \log\left(\frac{1}{\sqrt{2\pi}\sigma}\right)}_{\text{constant}} - \sum_{i=1}^{m} \frac{(y^{(i)} - X^{(i)}\beta)^2}{(2\sigma^2)}$ ↳ constant

So, Maximum likelihood estimation (MLE)

$\sum_{i=1}^{m} \frac{(y^{(i)} - X^{(i)}\beta)^2}{2}$ → Minimize it so, that $l(\beta)$ may be maximized

we can also write as $(Y - X\beta)^T (Y - X\beta)$

$\begin{bmatrix} y^{(1)} - h_\beta(x^{(1)}) \\ \vdots \\ y^{(m)} - h_\beta(x^{(m)}) \end{bmatrix}^T \begin{bmatrix} y^{(1)} - h_\beta(x^{(1)}) \\ \vdots \\ y^{(m)} - h_\beta(x^{(m)}) \end{bmatrix}$ $\quad (Y - h_\beta(X))^T (Y - h_\beta(X))$

$= \text{scalar} = \sum_{i=1}^{m} (y^{(i)} - h_\beta(x^{(i)}))^2$

So, $\text{cost} = \sum_{i=1}^{m} \frac{(y^{(i)} - x^{(i)}\beta)^2}{2}$

$\frac{\partial(\text{cost})}{\partial\beta} = \sum_{i=1}^{m} \frac{(y^{(i)} - x^{(i)}\beta)(-x^{(i)})}{2} = \sum \frac{(x\beta - y^{(i)})(x^{(i)})}{2}$

$\frac{\partial^2(\text{cost})}{\partial\beta^2} = \sum \frac{(x^{(i)})(x^{(i)})}{2} = \sum_{i=1}^{m} \frac{x^{2(i)}}{2} \Rightarrow +ve \neq 0$

So, curvature

So, $\frac{\partial^2(\text{cost})}{\partial\beta^2} = +ve$ always means upcurve & have as global local minima with is to minimize the cost that should be

as SO to maximize $l(\beta)$

So, cost becoming $\frac{\sum (y^{(i)} - h_\beta(x^{(i)}))^2}{2}$ is best such that global minima possible.

$Y = X\beta + \varepsilon$ $\quad h_\beta(x) = X\beta$

$J(\beta) = \frac{1}{2m} \sum (y^{(i)} - h_\beta(x^{(i)}))^2$

$J(\beta) = \frac{1}{2m} (Y - X\beta)^T (Y - X\beta)$

$J(\beta) \rightarrow$ scalar $\rightarrow$ average cost fun

[OLS] closed form OLS Normal Eqn

drive closed form parameter estimator $\hat{\beta}_{OLS}$

we can obtain our optimal β such that our $J(\beta)$ become minimized in single step $\Rightarrow \hat{\beta}_{OLS}$

$J(\beta) = \frac{1}{2m} (Y - X\beta)^T (Y - X\beta) = \frac{1}{2m} (Y^T - \beta^T X^T)(Y - X\beta)$

$J(\beta) = \frac{1}{2m} (Y^T Y - Y^T X\beta - \beta^T X^T Y + \beta^T X^T X\beta)$

$Y \in \mathbb{R}^{m \times 1}$, $Y^T \in \mathbb{R}^{1 \times m}$

$X \in \mathbb{R}^{m \times n}$, $X^T \in \mathbb{R}^{n \times m}$, $\beta^T \in \mathbb{R}^{1 \times n}$

So, $(\beta^T X^T Y)^T = (1 \text{ Dimen})^T$ $\quad \beta^T X^T Y \rightarrow (1 \times n)(n \times m)(m \times 1)$

$= 1 \text{ Dimen}$ $\quad \rightarrow$ scalar (1×1)

$= Y^T X\beta$

$J(\beta) = \frac{1}{2m} (Y^T Y - 2Y^T X\beta + \beta^T X^T X\beta)$

Date / Page

So in a we want to minimize the cost fn
$\nabla J_\beta = 0$ answer if $\nabla J_\beta \geq 0$
find β such that cost minimize

$$\nabla J_\beta = \frac{\partial J(\beta)}{\partial \beta} = \frac{1}{2m}(0 - 2X^TY + 2X^TX\beta) = \frac{1}{m}(X^TX\beta - X^TY)$$

$f(\beta) = Y^TX\beta \Rightarrow \nabla_\beta f = X^TY$ | $g(\beta) = \beta^TX^TX\beta$
$\partial f(\beta) = Y^TX\partial\beta$ | $\partial g(\beta) = \beta^TX^TX\partial\beta + (X\beta)^TX\partial\beta = (2X^TX\beta)^T\partial\beta$
$= (X^TY)^T\partial\beta$ | $\nabla g(\beta) = 2X^TX\beta$
$= (\nabla_\beta f)^T\partial\beta$

$$\nabla_2 J_\beta = \frac{\partial^2 J(\beta)}{\partial\beta^2} = \frac{1}{m}(X^TX) = H(\beta), \quad X \in \mathbb{R}^{m\times d},\ X^TX \in \mathbb{R}^{d\times m}\ X^TX \in \mathbb{R}^{d\times d}$$

$H(\beta)$ → hessian matrix

$(X^TX) = A \Rightarrow \beta^TA\beta = \beta^TX^TX\beta = (X\beta)^T(X\beta) = \|X\beta\|_2^2 \geq 0$

So, $H(\beta) = \frac{1}{m}(X^TX)$ will always be a semi positive definite matrix symmetric

it means that $\nabla_2 J(\beta) = H(\beta) \geq 0$ → upward bowl ∪

So, $\nabla J_\beta = 0 = \frac{1}{m}(X^TX\beta - X^TY) = 0$ will minimize

$X^TX\beta = X^TY$

$\hat{\beta}_{OLS} = (X^TX)^{-1}X^TY$ only if X^TX is full rank of (X^TX) else error

So, $\beta_{OLS} = (X^TX)^{\oplus}X^TY$

as well $H(\beta) = \frac{1}{m}X^TX = P\Lambda P^T$

$G^TP\Lambda P^TG = (P^TG)^T\Lambda(P^TG)$ $\quad \Lambda = diag(\lambda_0, \lambda_2, \ldots, \lambda_n)$

$= \sum_{i=0}^{n} \lambda_i \underbrace{\|P^TG\|_2^2}_{>0}$ so, $\lambda_i \geq 0$ as $G^THG \geq 0$

Date / Page

$X \xrightarrow{M} X^TX$ $\quad X \in \mathbb{R}^{m\times d}$

null space of $(X) \Rightarrow Xv = 0$ when $v \neq 0$

null space of $(X^TX) \Rightarrow X^TXv = 0$ when $v = 0$ $\quad X^TX \in \mathbb{R}^{d\times d}$

$v^TX^TXv = 0$

$(Xv)^T(Xv) = \|Xv\|^2 = 0$

only when $Xv = 0$ so,

nullspace of (X^TX) = null space (X)

rank + null space = no. of columns

$\Rightarrow$ rank(X) + null$(X) = d$, rank(X^TX) + null$(X^TX) = d$

So, rank(X) = rank(X^TX)

So, if $X \in \mathbb{R}^{m\times d}$ and $X^TX \in \mathbb{R}^{d\times d}$ $\quad d = n+1$

case I: X^TX is only invertible if its full rank $= d$

in X if $d > m \Rightarrow n \geq m$ then rank(X) capped at m which less then d

So, rank$(X^TX) < d$

So we can only use the code of pseudo-inverse non-invertible

$A = U\Sigma V^T$; $UU^T = I - U\bar{U}$; $V^TV = VV^T = I$

$A^{-1} = (U\Sigma V^T)^{-1} = (V^T)^{-1}\Sigma^{-1}U^{-1} = V\Sigma^{-1}U^T$

since $A^{-1}X$ then use $A^{\oplus}$

$A^{\oplus} = V\Sigma^{\oplus}U^T$ $\quad \Sigma^{\oplus}$ means $\forall \lambda_i$ if 0 then 0

case II: when features of X are collerelated meaning linear dependant on another column then also rank$(X) < d$ so is X^TX then (X^TX) non-invertible may use the concept of pseudo inverse

$\hat{\beta}_{OLS} = (X^TX)^{-1}X^TY$, $\nabla_\beta J(\beta) = \frac{1}{m} X^T(X\beta - Y)$

$(X^TX) \rightarrow$ noninvertible ① $d > m$ ② linear dependent features

use pseudo inverse (X^TX)

so, unique sol^n:

$$Rank(X^TX) = Rank(X) = d$$

[OLS] ③ OLD hessian & convexity conditions:

Strictly convex vs merely convex ? cost fun^n

hessian matrix $= \nabla^2_\beta J(\beta) = \frac{1}{m} X^TX = H$

$v^THv \geq 0$ — diagonalizable via Symmetric

semi PSD

$\lambda_i \geq 0$

so, strictly convexity of $J(\beta) \Rightarrow \lambda_i > 0$

merely convexity of $J(\beta) \Rightarrow \lambda_i \geq 0$

[OLS] ④ Orthogonal Projection & Hat Matrix Symmetry:

Define Projection hat matrix H & residual maker matrix $(I-H)$ & prove their symmetry, idempotency, orthogonality b/w col^m of space of X, the predicted vector $\hat{y}$ & residual vector e.

We know that $\hat{\beta}_{OLS} = (X^TX)^{-1}X^TY$

$$\hat{Y} = X\hat{\beta}_{OLS} = X(X^TX)^{-1}X^TY = HY$$

$$\hat{Y} = HY \quad ; H = X(X^TX)^{-1}X^T$$

H → Projection matrix

$X^TX \rightarrow$ square matrix

$H = X(X^TX)^{-1}X^T$

$H^2 = HH = X(X^TX)^{-1}(X^TX)(X^TX)^{-1}X^T = X(X^TX)^{-1}X^T = H$

$H^T = (X(X^TX)^{-1}X^T)^T = X(X^TX)^{-1}X^T = H$

so, Symmetric ($H = H^T$) And Idempotent ($H^2 = H$)

so, for Y we get HY

error $= e = Y - HY = (I-H)Y \Rightarrow (I-H)Y$

$e = Y - \hat{Y} = (I-H)Y$

$Y \rightarrow$ actual, $\hat{Y} \rightarrow$ predicted — $(I-H)$ residual maker matrix

$e \rightarrow$ residual error

$H = X(X^TX)^{-1}X^T$ → column space of input features (X)

$m \times n \quad n \times m \quad m \times m$ — $H \in \mathbb{R}^{m \times m}$

$m \rightarrow$ input features row

row space of X

$X = [1 \; X_1 \; X_2 \; \cdots \; X_n]$

$\hat{y} = HY$ = linear combo of row of $X \Rightarrow$ row space of X

residual $= y - \hat{y} = (I-H)Y$

(residual)$\cdot(\hat{y}) = ((I-H)Y)\cdot(HY)$

$= (H - H^2)Y^TY$

$= 0 \Rightarrow$ residual $\perp \hat{y}$

& $\hat{y}$ is col^m of X

so, ∴ proved



as well $(I-H)^T = (I-H)$

$(I-H)^2 = (I-H)$

$\hat{\beta} = (X^TX)^{-1}X^TY$

$\hat{Y} = X\hat{\beta} = X(X^TX)^{-1}X^TY = HY$

$H \rightarrow$ Projected (Hat) matrix such $H^2 = H$, $H = H^T$

there is a relation:

col^n space of X, Predicted vector $\hat{y}$ & residuals are orthogonal to each other

$R^m = Col(X) \oplus Null(X^T)$

$\hat{y} = X\hat{\beta}$ → linear combo of col^n of X

$\hat{y} \in Col(X)$

$e = y - X\hat{\beta} \Rightarrow X^Te = X^T(y - X\hat{\beta}) = X^Ty - X^TX\hat{\beta}$

as $\hat{\beta} = (X^TX)^{-1}X^Ty$

so, $X^Te = 0$

$e \in Null(X^T)$

But we know $Null(X^T) = (Col(X))^{\perp}$

so, $e \perp Col(X)$

$\hat{y} \in Col(X)$

$e \in Col(X)^{\perp}$

so, $\hat{y}^Te = 0$ $\hat{y} \perp e$

$R^m = Col(X) \oplus Null(X^T)$

OLS ⑤ Probabilistic Justification of OLS (Gaussian MLE): State the i.i.d Gaussian noise generative model for linear regression & prove why MLE of parameters equivalent to minimizing OLS squared error

✓ Proved

OLS ⑥ Gauss-Markov Statistical Properties of $\hat{\beta}_{OLS}$: Under standard linear model assumptions, check whether $\hat{\beta}_{OLS}$ is unbiased & derive its exact 2nd parameter covariance matrix

$Y = X\beta^* + \varepsilon$ ($\beta^* \rightarrow$ actual β) $\quad \varepsilon_i \sim N(0, \sigma^2)$; $\varepsilon = [\varepsilon_1\ \varepsilon_2 \cdots \varepsilon_m]^T$

Bias → When we calculate various $\hat{\beta}$ in our experiments then the estimate of all such various against the actual solution difference (bias)

$Bias(\hat{\beta}) = E[\hat{\beta}] - \beta^*$

$\hat{\beta}_{OLS} = (X^TX)^{-1}X^TY$

$E[\varepsilon] = 0$, $Var(\varepsilon) = \sigma^2$

$E[Y] = X\beta^*$

$Var(Y) = Var(X\beta + \varepsilon) = Var(\varepsilon) = \sigma^2 I$

$E[\hat{\beta}_{OLS}] = (X^TX)^{-1}X^TE[Y] = \beta^*$

$Bias(\hat{\beta}) = E[\hat{\beta}] - \beta^* = \beta^* - \beta^* = 0$

$\hat{\beta}_{OLS} = (X^TX)^{-1}X^TY = AY$

$Cov(\hat{\beta}_{OLS}) = Cov(AY) = A\,Cov(Y)\,A^T = \sigma^2 AA^T$

$Cov(Y) = Var(Y) = \sigma^2 I$

$AA^T = (X^TX)^{-1}X^TX(X^TX)^{-1} = (X^TX)^{-1}$

$Cov(\hat{\beta}_{OLS}) = \sigma^2(X^TX)^{-1}$

$X^TX = A$ → $d \times d$ symm square

$A = PDP^T \Rightarrow D = diag(\lambda_1, \cdots, \lambda_d)$

so, $A^{-1} = PD^{-1}P^T$

$X^TX \Rightarrow \lambda_1, \lambda_2 \cdots, \lambda_d$

$D = diag(\lambda_1, \lambda_2 \cdots, \lambda_d)$

$D^{-1} = diag(1/\lambda_1, 1/\lambda_2 \cdots, 1/\lambda_d)$

so,

$tr(Cov(\hat{\beta}_{OLS})) = \sigma^2\left(\frac{1}{\lambda_1} + \frac{1}{\lambda_2} \cdots + \frac{1}{\lambda_d}\right)$

When we get $\hat{\beta}_{OLS}$ under standard linear model assumptions, our $\hat{\beta}_{OLS}$ is unbaised.
Bais mean $E[\hat{\beta}_{OLS}] - \beta^*$ (actual).
$Cov(\hat{\beta}_{OLS}) = \sigma^2 (X^TX)^{-1}$ d x d matrix
$Y = X\beta^* + \varepsilon$
$\varepsilon_i \sim N(0, \sigma^2)$

[OLS] ④ Express variance & $Tr(cov(\hat{\beta}_{OLS}))$ in term of λ of X^TX prove how near-collinearity affects estimator variance and matrix conditioning.

Variance → how wildly our estimated weight $\hat{\beta}$ jump around when data changes slightly due to ε

$var(\hat{\beta}) = cov(\hat{\beta}) = \sigma^2 (X^TX)^{-1}$
$X^TX \rightsquigarrow$ eigenvalue $\rightsquigarrow \lambda_1, \lambda_2 \cdots, \lambda_d$
$(X^TX)^{-1} \rightsquigarrow 1/\lambda_1, 1/\lambda_2 \cdots, 1/\lambda_d$
$\sigma^2 (X^TX)^{-1} \rightsquigarrow \sigma^2/\lambda_1, \sigma^2/\lambda_2 \cdots, \sigma^2/\lambda_d$
$tr(cov(\hat{\beta})) =$ sum of eigenvalue $= \sigma^2 \left(\frac{1}{\lambda_1} + \frac{1}{\lambda_2} \cdots + \frac{1}{\lambda_d}\right)$

So, if in X a feature is near collinear to other feature then $\lambda \to 0$ (near zero) or 0 same then $1/\lambda \to \infty$ then estimator variance trace shot to ∞ which is very high and not a good state.

So, matrix condition (X feature matrix) is that no. columns/features are collinear or near collinear as to prevent variance shot.

$tr(cov(\hat{\beta})) = \sigma^2 (1/\lambda_1 + 1/\lambda_2 \cdots + 1/\lambda_d)$
eigenvalue of $X^TX \rightsquigarrow \lambda_1, \lambda_2 \cdots, \lambda_d$
or square singular value of $X \rightsquigarrow \lambda_1, \lambda_2 \cdots, \lambda_d$
$var(\hat{\beta}) = cov(\hat{\beta}) = \sigma^2 (X^TX)^{-1}$
if features collinear/near collinear $\lambda_i \to 0$ or 0 then variance shoot to ∞ then matrix conditioning → don't input/remove collinear/near collinear features
Perfect collinear → X^TX → singular
near-perfect-collinear → inverse exist but eigen value very small (smallest one), cause high variance

[OLS] ⑤ Unbaised Noise variance Estimator: State & justify the degrees of freedom correction for the unbaised estimator $\hat{\sigma}^2$ of the observation noise variance.

$y = X\beta + \varepsilon \qquad \varepsilon \sim N(0, \sigma^2 I)$
$\hat{\beta} = (X^TX)^{-1} X^TY = (X^TX)^{-1} X^TX \beta + (X^TX)^{-1} X^T\varepsilon$
$\hat{\beta} = \beta + (X^TX)^{-1} X^T\varepsilon$
$e = y - \hat{y} = y - X\hat{\beta} = y - X(X^TX)^{-1}X^Ty$
$HX = X(X^TX)^{-1}X^TX = X \qquad H = X(X^TX)^{-1}X^T$
$e = (I-H)y = (I-H)X\beta + (I-H)\varepsilon = (I-H)\varepsilon$
$RSS = e^Te = \varepsilon^T(I-H)^T(I-H)\varepsilon = \varepsilon^T(I-H)\varepsilon$
$(I-H)^T = (I-H)\,; (I-H)^2 = (I-H)$
$E[RSS] = E[\varepsilon^T(I-H)\varepsilon]$
identity $E[\varepsilon^TA\varepsilon] = Tr(A\,cov(\varepsilon))$ ($E[\varepsilon\varepsilon^T]$)
$E[RSS] = Tr((I-H)cov(\varepsilon))$
$= Tr((I-H)\sigma^2 I) = Tr(\sigma^2 I - \sigma^2 H)$

$E[RSS] = \sigma^2 \mathrm{Tr}(I-H)$

$= \sigma^2 (n-p)$

~~$H = H(H) = p \Rightarrow \mathrm{rank}(H) = p$~~

$E[RSS] = \sigma^2 (n-p)$

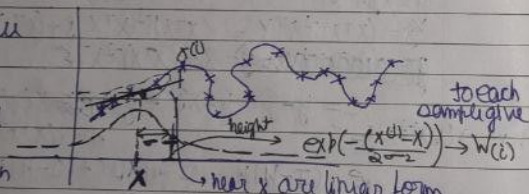
$E[\hat{\sigma}^2] = \sigma^2$?

$E[RSS] = \sigma^2 (n-p)$ ($n-p$)

$\hat{\sigma}^2 = \frac{RSS}{n-p} = \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{n-p}$

$\Rightarrow E[\hat{\sigma}^2] = E[\sigma^2]$

LLS ④ locally weighted linear Regression (LWLR) formulate distance weighted local regression cost fun for a query point x, derived its closed form matrix form eqn



Nonlinear

clearly look not linear eqn don't fit but when we look closely near its points

let, assume we give each of m $(x^{(i)}, y^{(i)})_{i=1}$ a weight $w_{(i)}$ $i=1$ to m such that $w_{(i)} \in [0,1]$

$w = \exp\left(-\frac{(x^{(i)} - x)}{2\sigma^2}\right)$ centered around x

$W \rightarrow$ shape of bell

$\Rightarrow J(\beta) = \frac{1}{2m} \times \sum_{i=1}^{m} w_{(i)} (x^{(i)}\beta - y)^2$

$\beta \rightarrow [\beta_0 \beta_1 \cdots \beta_n]^T$

$\cdot (Y - X\beta)^T (Y - X\beta)$

$\begin{bmatrix} y^{(1)} - x^{(1)}\beta \\ \vdots \\ y^{(m)} - x^{(m)}\beta \end{bmatrix}^T \begin{bmatrix} y^{(1)} - x^{(1)}\beta \\ \vdots \\ y^{(m)} - x^{(m)}\beta \end{bmatrix}$

$1 \times m \quad (n \times m)(m \times 1) \quad (1 \times m) \quad R^{1 \times m} \quad R^{m \times 1}$

$W = \mathrm{diag}(w_1, w_2, \ldots, w_m)$

$J(\beta) = \frac{(W(Y - X\beta))^T (Y - X\beta)}{2m} = \frac{(Y - X\beta)^T W (Y - X\beta)}{2m}$

$w_{(i)} = \exp\left(-\frac{(x^{(i)} - x)}{2\sigma^2}\right) \rightarrow$ independent of β

$\Rightarrow$ if $W \rightarrow$ independent of β $\quad w_{(i)} \in [0,1]$

$J(\beta) = \frac{1}{2m} \left((Y^T - \beta^T X^T)(WY - WX\beta)\right)$

$Y^T WY - Y^T WX\beta - \beta^T X^T WY + \beta^T X^T WX\beta$

$J(\beta) = \frac{1}{2m} \left(\underbrace{Y^T WY}_{0} \underbrace{- 2Y^T WX\beta}_{-2X^T WY} + \underbrace{\beta^T X^T WX\beta}_{2\beta^T X^T WX \partial\beta}\right)$

$(-2Y^T WX)^T$

$(2\beta^T X^T WX)^T$

$2X^T WX\beta$

$\Rightarrow \nabla_\beta J(\beta) = \frac{1}{m} \left(X^T WX\beta - X^T WY\right) = \frac{X^T W}{m} (X\beta - Y)$

$\nabla^2_\beta J(\beta) = \frac{X^T WX}{m}$ $\qquad \beta^T X^T WX \beta \geq 0$ (symm) $\Rightarrow Q \wedge Q^T$

$P Q \wedge Q^T P \Rightarrow P Q \wedge (PQ)^T$

$\Rightarrow$ ✓ minimum

$\Rightarrow X^T WX\beta = X^T WY$

$\beta = (X^T WX)^{-1} X^T WY$

$$\hat{\beta} = (X^T W X)^{-1} X^T W Y$$

where

$$W = \text{diag}(w_1, w_2, \dots, w_m) \quad W \in \mathbb{R}^{m \times m}$$

$$w_i = \exp\left(-\frac{(x - x^{(i)})^2}{2\tau^2}\right)$$

for locally weighted linear regression (LWLR)

$$\hat{\beta}_{\substack{\text{locally} \\ \text{LWLR}}} = (X^T W X)^{-1} X^T W Y$$

$$W = \text{diag}(w_1, w_2, \dots, w_m) \quad W \in \mathbb{R}^{m \times m}$$

provide weight to every sample $(x^{(i)}, y^{(i)})_{i=1}^{m}$

$$w_i = \exp\left(-\frac{(x - x^{(i)})^2}{2\tau^2}\right) \quad w_i \in [0, 1]$$

GD → Gradient descent

GD (1) Steepest Descent $\mathbb{R}^n$:

If we are at β and move $\Delta\beta$ then $J(\beta) \to J(\beta + \Delta\beta)$

since we want to minimize cost $J(\beta) > J(\beta + \Delta\beta)$

let define; $f(r) = J(\beta + r\Delta\beta)$

$f(0) = J(\beta)$, $f(1) = J(\beta + \Delta\beta)$

$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2}x^2 \dots$ (contd)

$f(r) = J(\beta + r\Delta\beta) = J(\beta_0 + \beta_1 \dots + \beta_n + r(\Delta\beta_1 + \Delta\beta_2 \dots + \Delta\beta_n))$

$\frac{\partial f(r)}{\partial r} = f'(r) = \dots \Rightarrow$

$\frac{\partial f'(r)}{\partial r} = f''(r) = \dots$

$\beta_j \to \beta_j + \Delta\beta_j$

$$f'(r) = \sum_{j=0}^{n} \frac{\partial J(\beta + r\Delta\beta)}{\partial \beta_j} \left(\frac{\partial \beta_j}{\partial r}\right) \Delta\beta_j$$

$$\nabla_\beta J = \begin{bmatrix} \partial J/\partial \beta_0 \\ \vdots \\ \partial J/\partial \beta_n \end{bmatrix}, \quad \Delta\beta = \begin{bmatrix} \Delta\beta_0 \\ \vdots \\ \Delta\beta_n \end{bmatrix}$$

$$f''(r) = (\nabla_\beta J)^T \Delta\beta = (\nabla_\beta J(\beta + r\Delta\beta))^T \Delta\beta$$

$$f''(r) = \sum_{j=0}^{n} \sum_{i=0}^{n} \frac{\partial^2 J(\beta + r\Delta\beta)}{\partial \beta_j \partial \beta_i} \Delta\beta_j \Delta\beta_i$$

$$H_{ij} = \frac{\partial J}{\partial \beta_j \partial i} = H_{ji} = \frac{\partial J}{\partial \beta_i \partial \beta_j}$$

$$f''(r) = \Delta\beta^T H \Delta\beta$$

so, $f'(0) = (\nabla_\beta J(\beta))^T \Delta\beta$, $f''(0) = \Delta\beta^T H \Delta\beta$

$\omega_i = \frac{v_i^2}{\sum v_i^2}$ $\quad \omega_i \geq 0, \sum_{i=1}^{n} \omega_i = 1$

$\frac{g^T H g}{g^T g} = \sum \omega_i \lambda_i$

So,

$$\lambda_{min} \leq \frac{g^T H g}{g^T g} \leq \lambda_{max} \quad \left[\frac{g^T g}{g^T H g} \in \left[\frac{1}{\lambda_{max}}, \frac{1}{\lambda_{min}}\right]\right]$$

at $Ag = \lambda_{min} g$ $\quad \frac{g^T g}{g^T g} = \lambda$

So, our case $\quad 0 < \alpha < \frac{2 g^T g}{g^T H g}$

Upper bound by $2/\lambda_{max}$

So, $\boxed{0 < \alpha < 2/\lambda_{max}}$

$\rightarrow J(\beta - \alpha g) = J(\beta) - \alpha g^T g + \frac{\alpha^2}{2} g^T H g$

$\frac{\partial J(\beta - \alpha g)}{\partial \alpha} = -g^T g + \alpha g^T H g = 0$

$\boxed{\alpha^* = \frac{g^T g}{g^T H g}}$

Exact line search

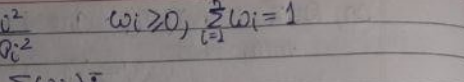
$\alpha \rightarrow$ parameter optimal at every step

But works good only if no. of features are small else too many calculation

$\nabla_\beta^2 J = H = \frac{1}{m} X^T X$

(height)

* at contour surface $J(\beta)$ is same



$J(S) = J(\beta)$

$J'(S) = 0$ as per each

at β on surface & move $\Delta\beta$ along tangent

$J(\beta) - J(\beta + \Delta\beta) = J(S) = J = \text{Constant}$

So, $\frac{\partial J(\beta(S))}{\partial S} = 0$

$\sum \frac{\partial J(\beta(S))}{\partial \beta_j} \frac{\partial \beta_j(S)}{\partial S} \Rightarrow (\nabla_\beta J(\beta))^T \text{(tangent)} = 0$

$\text{tangent} = \begin{bmatrix} \partial \beta_0(S)/\partial S \\ \partial \beta_n(S)/\partial S \end{bmatrix}$ $\quad \cos\theta = 90$ So,

gradient $\perp$ to contours of surface

So, to minimize our cost fun $J(\beta)$ at every step its necessary that our $\Delta\beta$ step which is $\alpha u \rightarrow (\alpha > 0, \|u\|_2 = 1)$ becomes

$\Delta\beta = \text{step} = -\alpha \nabla_\beta J(\beta) = -\alpha \nabla_\beta J(\beta)$

-ve gradient points in the steepest local decrease

[GD] ① Batch Gradient Descent (BGD) for OLS:

Update rule (scalar & matrix form)

On cost fun OLS & state its per iteration time complexity.

$$J(\beta) = \frac{1}{2m}\left(\sum_{i=1}^{m} (y^{(i)} - h_\beta(x^{(i)}))^2\right)$$

$(x^{(i)}, y^{(i)})_{i=1}^{m}$

$x^{(i)}\beta = \beta_0 + x_1^{(i)}\beta_1 + \cdots + x_n^{(i)}\beta_n$

$\beta_j := \beta_j - \alpha \frac{\partial J}{\partial \beta_j}$

$$\frac{\partial J(\beta)}{\partial \beta_j} = \frac{1}{2m} \times \left(\sum_{i=1}^{m} (y^{(i)} - x^{(i)}\beta)(-x_j^{(i)})\right)$$

$$\frac{\partial J(\beta)}{\partial \beta_j} = \frac{1}{m}\sum_{i=1}^{m} (x^{(i)}\beta - y^{(i)})(x_j^{(i)})$$

$$\beta_j := \beta_j - \frac{\alpha}{m}\sum_{i=1}^{m} (x^{(i)}\beta - y^{(i)})(x_j^{(i)})$$

$X^T(X\beta - Y)$

$$\beta := \beta - \frac{\alpha}{m} X^T(X\beta - Y)$$

$$\begin{bmatrix} 1 & \cdots & 1 \\ x_{11} & \cdots & x_{m1} \\ \vdots & & \vdots \\ x_{1n} & \cdots & x_{mn} \end{bmatrix} \begin{bmatrix} x^{(1)}\beta - y^{(1)} \\ \vdots \\ x^{(m)}\beta - y^{(m)} \end{bmatrix}$$

we repeat

{

$\beta_j := \beta_j - \frac{\alpha}{m}\sum_{i=1}^{m} (x^{(i)}\beta - y^{(i)})(x_j^{(i)})$; for all j

} till our loss stops decreasing

during a single epoch (full train data) result in β update only one time.

Advantage is that once we reach a optimal state such that $\sum (x^{(i)}\beta - y^{(i)}) x_j^{(i)}$ become zero or averages out to zero then it remains zero & β remain same.

repeat {

$\beta_j := \beta_j - \frac{\alpha}{m}\sum_{i=1}^{m} (x^{(i)}\beta - y^{(i)})(x_j^{(i)})$; for all j $j = 0, \cdots, n$

(or)

$\beta := \beta - \frac{\alpha}{m} X^T(X\beta - Y)$ (BGD)

} till loss stops decreasing

$\equiv$ slow for large m

during a single epoch → β update only one time can reach global minimum & retain there & don't move. $\beta = [\beta_0\ \beta_1 \cdots \beta_n]^T$

SGD ③ Stochastic Gradient Descent (SGD) for OLS:

single sample SGD update

$\beta_j := \beta_j - \alpha (x^{(i)}\beta - y^{(i)})(x^{(i)})$

repeat {

In this for each m {

$\beta_j := \beta_j - \alpha (x^{(i)}\beta - y^{(i)})(x^{(i)})$; for each j

}

} till loss stops decreasing

although this process update β m times in each epoch. This is a fast process. But in this our β can never remain at global minimum of $J(\beta)$ but stuck or reach near the lowest possible global min then asymptotic as our term is just $(x^{(i)}\beta - y^{(i)})(x^{(i)})$ for single sample rather then summation/average which can zero them out.

K → Kth batch, fixed batches

repeat {
  For K in 1 to m/b {
    $\beta_j := \beta_j - \frac{\alpha}{b} \sum_{i=1}^{b} (x^{k(i)}\beta - y^{k(i)})(x^{k(i)})$
  }
}
Mini-BGD

3 till our loss stops decreasing

1 epoch → β update m/b time
b → batchsize
if b = 1 ⇒ SGD; b = m ⇒ BGD
1 < b < m ⇒ trade b/w gradient variance and the speed of calculation
SGD → highest variance, fast
BGD → zero variance, slow

GD (5) Exact line search (Cauchy step size): derive the closest form optimal step size α_t^* that exactly minimizes a quadratic objective along the current negative gradient direction at iteration t

Exact Line search

$$\alpha_t^* = \frac{g^T g}{g^T H g}$$

where α_t^* calculated at every t iteration such that

changes with iterations → $g = \nabla_\beta J(\beta)$

remain same → $H = \frac{1}{m} X^T X$

so only calculate ones time

GD (6) Fixed-step BGD error recursion & Spectral Stability Bound: derive recurrence relation for parameter error vector $e^{(t)} = \beta^{(t)} - \beta^*$ under fixed step G.D. batch λ. give upper & lower bound of α require for convergence

$$e^{(t)} = \beta^{(t)} - \hat{\beta}$$

$$\hat{\beta} = (X^T X)^{-1} X^T Y \qquad 1/m X^T X = H$$

$$J(\beta) = \frac{1}{2m} (X\beta - Y)^T (X\beta - Y)$$

$$\nabla J(\beta) = \frac{1}{m} X^T (X\beta - Y) = \frac{1}{m} X^T X \beta - \frac{1}{m} X^T Y$$

$$\frac{X^T X}{m} \hat{\beta} = \frac{X^T Y}{m} = H\hat{\beta} \Rightarrow \nabla J(\beta) = H\beta - H\hat{\beta}$$

$$\nabla J(\beta^{(t)}) = H(\beta^{(t)} - \hat{\beta}) = H e^{(t)}$$

$$\beta^{(t+1)} = \beta^{(t)} - \alpha \nabla J(\beta^{(t)})$$

$$\beta^{(t+1)} - \hat{\beta} = \beta^{(t)} - \hat{\beta} - \alpha \nabla J(\beta^{(t)})$$

$$e^{(t+1)} = e^{(t)} - \alpha H e^{(t)} = (I - \alpha H) e^{(t)}$$

$H = 1/m X^T X$ → square symmetric matrix $H = P \Lambda P^T$; $P P^T = P^T P = I$

$$(I - \alpha H) = (P I P^T - P(\alpha\Lambda) P^T) = P(I - \alpha\Lambda) P^T$$

$\Lambda = \text{diag}(\lambda_0, \lambda_2, \ldots, \lambda_n)$ · λ_i → eigenvalue of H

$I - \alpha\Lambda = \text{diag}(1 - \alpha\lambda_0, 1 - \alpha\lambda_1, \ldots, 1 - \alpha\lambda_n)$

$e^1 = (I - \alpha H) e^0$

$e^2 = (I - \alpha H)^2 e^0$

$e^{(t+1)} = (I - \alpha H)^t e^0$

as $t \to \infty$, we want our error of parameter $= \beta^\infty - \hat{\beta} \to 0$ converge

$e^\infty = (I - \alpha H)^\infty e^0$

$e^0 \neq 0$

So, for error to converge to 0
it necessary from $(I-\alpha H)^{\infty}=0$

$$Q(I-\alpha\Lambda)^{\infty}Q^{T}=0$$
$$(I-\alpha\Lambda)^{\infty}=0$$

So, means
$(1-\alpha\lambda_n)^{\infty}=0$ for each n to n

means
$|1-\alpha\lambda_n|<1$

$-1<1-\alpha\lambda_n<1 \Rightarrow -2<-\alpha\lambda_n<0$

$0<\alpha\lambda_n<2$

$$0<\alpha<\frac{2}{\lambda_n}$$

$\lambda = \lambda_{min}$ to λ_{max}

So,
$$\boxed{0<\alpha<\frac{2}{\lambda_{max}}}$$

✓ ① If $(1/\lambda_{max} < \alpha < 2/\lambda_{max}) \Rightarrow$ + to − to + ... zig-zag shape convergence

✓ ② If $(\alpha = 1/\lambda_{max}) \Rightarrow$ single step

✓ ③ If $(0<\alpha<1/\lambda_{max}) \Rightarrow$ remain the same, slowly crawl to global min

~ ④ If $(\alpha = 2/\lambda_{max}) \Rightarrow$ oscillate between remain as

X ⑤ Else case $\Rightarrow$ diverges

$e^{(t)} = \beta^{(t)} - \hat{\beta}$ = error of parameter

recursion relation $\Rightarrow e^{(t+1)} = (I-\alpha H)e^{(t)}$

So, $e^{(t)} = (I-\alpha H)^{t} e^{0}$

$H \rightarrow$ square symmetric $= Q\Lambda Q^{-1}$ $\Lambda = \lambda_0, \lambda_1 \dots, \lambda_n$

$(I-\alpha H) = Q(I-\alpha\Lambda)Q^{T}$

condition of convergence each $\lambda_i \equiv (1-\alpha\lambda_i)^{\infty}$

as $e^{\infty} \rightarrow 0$ as $t \rightarrow \infty$ then $|1-\alpha\lambda_i| < 1$

So, $0<\alpha<\frac{2}{\lambda_{max}}$

for zig-zag if $1/\lambda_{max} < \alpha < 2/\lambda_{max}$ — zig-zag convergence takes a lot of time

So, $0<\alpha<1/\lambda_{max}$ for crawl convergence — better

[GO] ⑦ Optimal Fixed learning Rate & Condition Number Convergence Rate : derive single optimal fixed step size α^* for quadratic loss & express worst case per step error contraction factor in terms of Hessian condition no.

In Hessian Matrix : $H \Rightarrow \lambda_i$ eigenvalues

$$K(H) = \frac{\lambda_{max}}{\lambda_{min}}$$

$e^{(t)} = \beta^{(t)} - \hat{\beta}$

$e^{(t)} = (I-\alpha H)^{(t)} e^{(0)}$

$e^{(t)} = Q(I-\alpha\Lambda)^{(t)} Q^{T} e^{(0)}$

$H = Q\Lambda Q^{T}$

$\lambda_i \rightarrow$ eigenvalue

Eigenvector creation $\Rightarrow$ as

$$z_{i,k+1} = (1-\alpha\lambda_i) z_{i,k}$$

for convergence we need $0<\alpha<\frac{2}{\lambda_{max}}$

Worst case contraction factor per step | largest of these magnitudes tells how much error can shrink at worst

$$\rho(\alpha) = \max_i |1 - \alpha\lambda_i|$$

→ Optimal fixed learning rate:

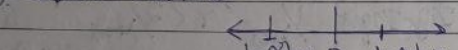
to minimize the worst case contraction

$$\min_\alpha \max_i |1 - \alpha\lambda_i|$$

$\lambda_i \in [\lambda_{min}, \lambda_{max}] \Rightarrow -\alpha\lambda_i \in [-\alpha\lambda_{max}, -\alpha\lambda_{min}]$

$1 - \alpha\lambda_i \in [1 - \alpha\lambda_{max}, 1 - \alpha\lambda_{min}]$

So, choose α such that



so, that $|1 - \alpha\lambda_{max}| = |1 - \alpha\lambda_{min}|$

optimal point: symmetrical around 0

So, that we can minimize the worst case spread

so

$1 - \alpha\lambda_{max} = \alpha\lambda_{min} - 1$; 0 1 $1-\alpha\lambda_{min}$

$2 = \alpha(\lambda_{max} + \lambda_{min})$

$$\alpha^* = \frac{2}{\lambda_{max} + \lambda_{min}}$$

then

optimal contraction factor

$$\rho(\alpha^*) = \left|1 - \frac{2\lambda_{max}}{\lambda_{min} + \lambda_{max}}\right| = \frac{\lambda_{max} - \lambda_{min}}{\lambda_{max} + \lambda_{min}}$$

$$\rho^* = \frac{\lambda_{max} - \lambda_{min}}{\lambda_{max} + \lambda_{min}} = \frac{\kappa(\kappa) - 1}{\kappa(\kappa) + 1} \approx 1 - \frac{2}{\kappa}$$

$\kappa \approx 1$ ⇒ convergence very fast

$\kappa \gg 1$ ⇒ ill conditioned ⇒ Error decrease very slowly

smaller ρ ⇒ faster convergence

$\rho = 0 \Rightarrow \kappa = 1 \Rightarrow$ one-step

Optimal $\rho^* = \frac{\kappa - 1}{\kappa + 1}$

α → how big step we take

ρ → how much error survives after that step

For a quadratic fun $J(\theta)$ & fixed learning rate (α) G.D, the theoretical optimal constant step size is

$$\alpha^* = \frac{2}{\lambda_{max} + \lambda_{min}}$$

worst case contraction factor per step ($\rho(\alpha)$)

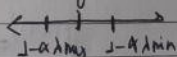
(largest of these magnitudes tell how much error can shrink at worst)

$$\rho(\alpha) = \max_i |1 - \alpha\lambda_i|$$

So, optimal fixed α: minimize worst case contradiction

$$\min_\alpha \left(\max_i |1 - \alpha\lambda_i|\right)$$

So, $|1 - \alpha\lambda_{max}| = |1 - \alpha\lambda_{min}|$

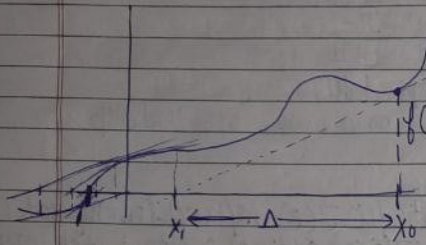


Optimal contradiction factor: $\rho(\alpha^*) = \rho^* = \frac{\lambda_{max} - \lambda_{min}}{\lambda_{max} + \lambda_{min}}$

$$\rho^* = \frac{\kappa - 1}{\kappa + 1} \qquad \kappa = \frac{\lambda_{max}}{\lambda_{min}}$$

$\kappa = 1 \Rightarrow$ convergence fast

[60] ⑧ Notwton-Methord for Optimization



$x_0 = x_1 + \Delta$
$x_1 = x_0 - \Delta$

$f'(x_0) = \frac{f(x_0)}{\Delta}$

$\Delta = \frac{f(x_0)}{f'(x_0)}$

$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$

$x^{(t+1)} = x^{(t)} - \frac{f(x^{(t)})}{f'(x^{(t)})}$

~~to~~ find $J'(\beta) = 0$ to β

$\beta^{(t+1)} = \beta^{(t)} - \frac{J'(\beta^{(t)})}{J''(\beta^{(t)})} = \beta^{(t)} - (H)^{-1} \nabla_\beta J(\beta)$

$\beta^{(t+1)} = \beta^{(t)} - (H)^{-1} \nabla_\beta J(\beta^{(t)})$

$H = \nabla^2 J(\beta^{(t)})$ ⊕

$X^T X = d \times d \quad d \times 1 = d \times 1$

$H = \frac{1}{m} X^T X$, $\nabla_\beta J(\beta) = \frac{1}{m} X^T (X\beta - y)$

$H^{-1} = m(X^T X)^{-1}$

$(H^{-1}) \nabla_\beta J(\beta) = \beta - (X^T X)^{-1} X^T Y$

$(H^{-1}) \nabla_\beta J(\beta) = \beta - (X^T X)^{-1} X^T Y$

So, $\beta^{(t+1)} = \beta^{(t)} - \beta^{(t)} + (X^T X)^{-1} X^T Y$

$\beta^{(t+1)} = (X^T X)^{-1} X^T Y$

exactly equal to β_{OLS}

OLS Objective perfectly quadratic

$\beta^{(1)} = (X^T X)^{-1} X^T y$

So, converges in exactly 1 iteration from start point, assume $X^T X$ invertible

let $\beta^* = (X^T X)^{-1} X y$

$\beta_{n+1} = \beta^*$ independent of β_n

So, after one netwon step $e_{n+1} = 0$

1-step convergence

BGD : m: samples, n: features

$\nabla J = X^T (X\beta - y)$ } $O(mn)$ cost per iteration

$O(mn)$ $\quad O(mn)$

newton: $H = \frac{1}{m} (X^T X)$ $O(mn^2)$ cost

inverse $O(n^3)$

step $H^{-1} g$ $O(n^2)$

Overall $O(mn^2 + n^3)$ 1-step OLS netwon

BGD : $\beta_{n+1} = \beta_n - \alpha \nabla J$ ⇒ many iteration

Newton: $\beta_{n+1} = \beta_n - H^{-1} \nabla J$ ⇒ 1-iteration in case of OLS

Newton-Raphson method:
find $f(x)=0$ then x

then $\beta^{(t+1)} = \beta^{(t)} - \frac{J'(\beta)}{J''(\beta)} = \beta^{(t)} - \left(\frac{\partial^2 J}{\partial \beta \partial \beta^T}\right)^{-1} \frac{\partial J}{\partial \beta}$

$\beta^{(t+1)} = \beta^{(t)} - (H)^{-1} \nabla_\beta J(\beta)$

in OLS condition $\Rightarrow$

$\beta^{(t+1)} = \beta^* = (X^TX)^{-1}X^TY$

1-step convergence $e_{n+1}=0$

Cost of newton $\Rightarrow O(mn^2+n^3)$

BGD $\Rightarrow O(mn)$

Condition no. $K(H) = \frac{\lambda_{max}}{\lambda_{min}} \geq 1$

at $\hat{\beta}$ we have $J(\beta)$ $\nabla J(\hat{\beta}) = 0$

$J(\beta) = J(\hat{\beta}) - \alpha 0 + \frac{\alpha^2}{2}(\beta-\hat{\beta})^T H (\beta-\hat{\beta})$

$\beta - \hat{\beta} = C \Rightarrow J(\beta) - J(\hat{\beta}) = \frac{C^T H C}{2} = \frac{C Q \Lambda Q^T C}{2}$

$H = Q \Lambda Q^T$

at contours $J(\beta) - J(\hat{\beta}) =$ constant $= C$

$\alpha C = Z^T \Lambda Z = C'$

$\lambda_0 v_0^2 + \lambda_1 v_1^2 \cdots + \lambda_n v_n^2 = C'$

$\frac{v_0^2}{C/\lambda_0} + \frac{v_1^2}{C/\lambda_1} \cdots + \frac{v_n^2}{C/\lambda_n} = 0$

so, ellipse axis of contours $\alpha 1/\sqrt{\lambda_i}$

steepest / smallest axis $\Rightarrow \lambda_{max}$ along

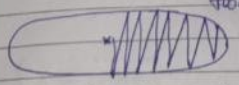
longest axis $\Rightarrow \lambda_{min}$ along

$K(H) = \frac{\lambda_{max}}{\lambda_{min}}$ large then error reduction per step also very small

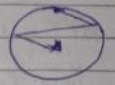
so, $x_j^{(i)} \leftarrow \frac{x_j^{(i)} - \mu_j}{\sigma_j}$ (normalize) so, that range of circle shape & prevent zigzag

normalize $[0,1]$

too many steps



on little steps



if $\eta_{max} \approx 10^6$ then itself look large

$\eta_{min} \approx 1$

circle when $\lambda_{max} = \lambda_{min}$

(Error along axis at step t) $= (1-\lambda_i \alpha)^t$ (initial error along axis)

for $H = 1/m\, X^TX$ our eigenvalue $\lambda_0, \cdots \lambda_n$ those are the contours ellipse / circle with axis stretch $\alpha 1/\sqrt{\lambda_i}$

shortest axis $\Rightarrow 1/\sqrt{\lambda_{max}}$ } $\rightarrow$ elliptical shape

longest axis $\Rightarrow 1/\sqrt{\lambda_{min}}$ }

for circle $\Rightarrow \lambda_{min} = \lambda_{max}$

elliptical $\Rightarrow$ can have zig-zag issues even when $\alpha < 2/\lambda_{max}$ GD $\nabla J(\beta)$ don't walk to $\hat{\beta}$ path and takes a lot of step to converge $K \gg 1$ as $\lambda_{max} \gg \lambda_{min}$. so, if we normalize each feature so, that step can decrease

$x_j^{(i)} \rightarrow \frac{x_j^{(i)} - \mu_j}{\sigma_j}$ $x_j^{(i)} \sim [0,1]$

Error along axis at step t $= (1-\alpha\lambda_i)^t$ (initial error along axis)

Let S be scaling diagonal matrix

$S = \text{diag}(c_1, c_2 \cdots, c_d)$

$X_{new} = XS$ scalled features

$\hat{\beta}_{newOLS} = (X_{new}^T X_{new})^{-1} X_{new}^T Y$

$= (SX^T XS)^{-1} SX^T Y$

$= S^{-1}(X^T X)^{-1} S^{-1} SX^T Y$

$\hat{\beta}_{newOLS} = S^{-1} \hat{\beta}_{OLDOLS}$

$\hat{Y}_{new} = X_{new} \hat{\beta}_{newOLS} = XSS^{-1} \hat{\beta}_{OLDOLS} = X\hat{\beta}_{OLDOLS}$

$Y_{new} = Y_{OLD} = Y$

not change

After alwe normalize features the output doesn't change.

we normalize/scale features

$S = \text{diag}(c_1, c_2 \cdots, c_d)$

$X_{new} = XS$

$\hat{\beta}_{newOLS} = S^{-1} \hat{\beta}_{OLDOLS}$

$Y_{new} = Y_{OLD} = Y$

$L_1 \rightarrow$ Lasso Regularizations
$L_2 \rightarrow$ Ridge

[L1L2] ① Ridge (L_2) regression objective & closed-Form Normal eq^n : Boudgi

Why need for regularization:

$\hat{\beta}_{OLS} = (X^T X)^{-1} X^T Y$

if in X $m < d$ @ features calculated highly making $\lambda_i \rightarrow 0$ or ≈ 0

then $X = U \Sigma V^T$ $\quad VV^T = I$

$X^T = V \Sigma^T U^T$

$X^T X = V \Sigma^T \Sigma V^T \sim V \Lambda V^T$

such that $\Lambda = \text{diag}(\lambda_1^2, \lambda_2^2 \cdots \lambda_d^2)$

if $\lambda_i \rightarrow 0$ $\quad (X^T X)^{-1} \rightarrow (V \Lambda V^T)^{-1} = V \Lambda^{-1} V^T$

not exact

then inverse exist

$\Lambda^{-1} = \text{diag}(1/\lambda_1^2 \cdots, 1/\lambda_d^2)$

$(X^T X)^{-1} X^T = V \Lambda^{-1} V^T V \Sigma^T U^T$

$= V (\Sigma^T \Sigma)^{-1} \Sigma^T \Sigma \Sigma^{-1} U^T$

$= V \Sigma^{-1} U^T$

∞, basically if our $\lambda_i \rightarrow 0$ features highly correlated then it make our $1/\lambda_i \rightarrow \infty$ which make our $\hat{\beta}_{OLS}$ also explode to ∞ as well if features are linear dependent on another $\lambda_i = 0$ in that case $(X^T X)^{-1}$ doesn't even exist forced to used pseudo inverse / causes loss of info.

② Unregularized : $J(\beta) = \frac{1}{2m} \sum (h_\beta(x^{(i)}) - y^{(i)})^2$

where

$J(\beta) = \frac{1}{2m} (X\beta - Y)^T (X\beta - Y)$

where

$\beta = [\beta_0, \beta_1 \cdots, \beta_n]^T$ $\quad X = [1 \; x_1 \cdots x_n]$

$1 \rightarrow$ col^m of 1's

Date / Page

We know that

if we have a univariate with no intercept

$Y = \beta X + \varepsilon$ — $X \to$ 1 feature

$J(\beta) = \frac{1}{2m} \sum_{i=1}^{m} (X\beta - Y)^2$ — $X = [x_1]$, $\beta = [\beta]$ not intercept

$\frac{\partial J(\beta)}{\partial \beta} = \frac{2}{2m} \sum_{i=1}^{m} (X\beta - Y) X^{(i)}$

$= \frac{1}{m} \sum_{i=1}^{m} \left((x^{(i)})^2 \beta - x^{(i)} y^{(i)} \right) = 0$

$\beta \sum_{i=1}^{m} (x^{(i)})^2 = \sum x^{(i)} y^{(i)}$

$\beta = \frac{\sum x^{(i)} y^{(i)}}{\sum (x^{(i)})^2} = \frac{\langle X, Y \rangle}{\langle X, X \rangle}$ regress Y on X

$\langle X, Y \rangle$ = dot product

$\langle X, Y \rangle = x^{(1)} y^{(1)} + x^{(2)} y^{(2)} + \dots + x^{(m)} y^{(m)}$ $X = [x^{(1)}, x^{(2)} \dots, x^{(m)}]$, $Y = [y^{(1)}, y^{(2)} \dots, y^{(m)}]$

$\langle X, X \rangle = (x^{(1)})^2 + \dots + (x^{(m)})^2$

if we have univariate with intercept then:

$Y = \beta_1 X_1 + \beta_0 + \varepsilon = X\beta + \varepsilon$ $X = [1 \; X_1]$, $\beta = [\beta_0 \; \beta_1]$

$\beta_0 \Rightarrow$ regress Y on $\bar{1}$

$\beta_0 = \frac{\langle Y, 1 \rangle}{\langle 1, 1 \rangle} = \frac{1}{m} \sum_{i=1}^{m} y^{(i)} = \bar{y}$

residual of $X_1 = X_1 - (\bar{x}_1)\bar{1} = \gamma_1$ that is orthogonal to all previous feature cols (I)

now; $\beta_1 \Rightarrow$ regress Y on γ_1

$\beta_1 = \frac{\langle Y, \gamma_1 \rangle}{\langle \gamma_1, \gamma_1 \rangle}$

Date / Page

So, from before this we can be sure even in case of our multivariate case + intercept our β_0 will remain $\bar{y}$

so, $\hat{\beta}_0 = \bar{y} = \frac{1}{m} \sum_{i=1}^{m} y^{(i)}$ (intercept to parameter)

So, we try to regularize / add penality to our cost fn^n if we change β_0 the corresponding y should also change that is what we don't want. So, our intercept remains unpenalized.

R → ridge

$J(\beta)^R = \frac{1}{2m} \left(\sum_{i=1}^{m} (x^{(i)}\beta - y^{(i)})^2 + \lambda \sum_{j=1}^{n} \beta_j^2 \right)$

$\lambda \to$ regularization parameter

also seen as $\sum_{j=1}^{n} \beta_j^2 \leq S$; mean our total buget of $\sum_{j=1}^{n} \beta_j^2$ can never exceed S

$J(\beta^R) = \frac{1}{2m} \left((X\beta - Y)^T (X\beta - Y) + \lambda \beta^T \Lambda \beta \right)$

$\beta^T \Lambda \beta$

$2\beta^T \Lambda \partial\beta = (2\Lambda\beta)^T \partial\beta$

$\Lambda = \text{diag}(0, 1, 1, \frac{1}{j}, \dots 1)$ — $n \times n$ matrix; 0 as we don't regularize intercept

$\nabla J(\beta) = \frac{\partial J(\beta^R)}{\partial \beta} = \frac{1}{2m} \left(2X^T (X\beta - Y) + 2\lambda \Lambda \beta \right)$

$X^T (X\beta - Y) + \lambda \Lambda \beta = 0 = X^T X\beta - X^T Y + \lambda \Lambda \beta$

$(X^T X + \lambda \Lambda)\beta = X^T Y$

$\hat{\beta}_{Ridg} = (X^TX + \lambda\Lambda)^{-1}X^TY$

closed form sol^n where $\Lambda = \text{diag}(0,1,1,\ldots,1)$ $d \times d$ matrix

$X^TX \Rightarrow$ symmetric square $\Rightarrow$ $X^TX = QDQ^T$

$Q^T(X^TX)Q$

$Q^TQ\,D\,Q^TQ$

V^TDV

$e_i \rightarrow$ eigenvalue

$\sum \lambda_i \|v\|_2^2 \geq 0$ so, $e_i = 0$ possible

so, inverse don't exist

further $X^TX + \lambda\Lambda = QDQ^T + \lambda\Lambda$

$V^T(QDQ^T)V + V^T\lambda\Lambda V > 0$

≥ 0 $\quad$ > 0

So, full terms eigenvalue > 0

also seen as $(X^TX)V = \lambda V$

$V \neq 0$

$(X^TX + \lambda\Lambda)V = (X^TX)V + \lambda\Lambda V$

$= \lambda V + \lambda V$ $\quad$ $\lambda \geq 1$, nearly $\Lambda = I$

$= (e + \lambda)V$

even if $e_i = 0$ but $\lambda > 0$

then $e_i + \lambda = \lambda > 0$

So, positive definite

Hence the out Inverse surely exist

or to be more precise

$V^TX^TXV + V^T\lambda\Lambda V$; $V \neq 0$

$\|XV\|_2^2 + \lambda\|V\|_2^2$ $\quad$ $V = [V_0\ V_R]$

$V_0 = 0$ $\quad$ $V_R = [V_1 \ldots V_n]$

$XV = 0$ $\quad$ $V_R = 0$ $\rightarrow$ $V = [V_0\ \bar{0}]$; $\bar{0}$ row 0

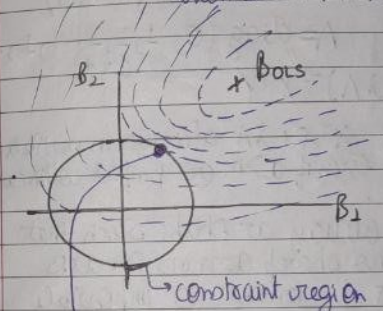
$X = [1\ x_1 \ldots x_n]$ $\quad$ $V = [V_0\ \bar{0}]$

if $V_0 = 0$ then $V = 0$ but contradict $V \neq 0$

So, $V^TX^TXV + V^T\lambda\Lambda V > 0$ always

So, $(X^TX + \lambda\Lambda)$ is always invertible.

Even when in X =) $d > n$ or features are Perfectly collinear



Suppose $\beta_j = 2$

$\beta_1^2 + \beta_2^2 \leq 5$

if OLS outside it

this is the first point of contact b/w the constraint region & our elliptical contours around OLS which will be the minimum $J(\beta)$ we can reach under constraint, if we take latter Point of contact that won't minimize $J(\beta)$ as much as possible as $J(\beta)$ increases as our elliptical contours axis increases

$(= \sum \lambda_i \beta_i^2$

Cost fun for Ridge (L_2) Regression is $\lambda > 0$

$J(\beta)_R = \frac{1}{2m}\left(\sum_{i=1}^{m}(X^{(i)}\beta - y^{(i)})^2 + \lambda\sum_{j=1}^{n}\beta_j^2\right)$

$\beta_0 = \sum y_i = \bar{y}$ (regress y on $\bar{1}$) (unpenalized intercept β_0)

$\beta_j = \frac{\langle Y, r_j \rangle}{\langle r_j, r_j \rangle}$ $\quad$ $\langle x, x \rangle \rightarrow$ dot product

regress Y on γ_j such γ_j is orthogonal com to previous all calculated

$\gamma_j = x_j - \sum \beta_n \gamma_n$

$$\gamma_j = X_j - \sum_{k=1}^{j-1} \gamma_k \beta_k \qquad ; \beta_k = \frac{\langle \gamma_k, X_j \rangle}{\langle \gamma_k, \gamma_k \rangle}$$

↓ orthogonal to $\gamma_0, \gamma_1, \cdots, \gamma_{j-1}$ $\quad \gamma_0 = I$ (intercept col)

$$J(\beta) = \frac{1}{2m}\left((X\beta - Y)^T(X\beta - Y) + \lambda \beta^T \Lambda \beta\right)$$

$$\Lambda = \text{diag}(0, 1, 1, \cdots, 1, 1) \quad (d \times d)$$

$$\hat{\beta}_{ridge} = (X^TX + \lambda\Lambda)^{-1} X^T Y$$

Advantage $X^TX + \lambda\Lambda$ always invertible even if $d > m$ or perfect collinear features

our $\sum_{j=1}^{d} \beta_j \leq S$ always a shape such that no sharp corners on axis that makes β_j becoming 0 impossible unless $\beta_j = 0$.
So, never reduces $\beta_j = 0$ only $\beta_j \to 0$

[4.4.2] ② BGD, SGD & MiniBGD for Ridge Regression (Weight decay):

$$\beta_j := \beta_j - \alpha \frac{\partial J}{\partial \beta}$$

BGD

repeat {

$$\beta_0 := \beta_0 - \frac{\alpha}{m} \sum_{i=1}^{m} (x^{(i)}\beta - y^{(i)}) x_0^{(i)}$$

$$\beta_j := \beta_j - \frac{\alpha}{m} \sum_{i=1}^{m} (x^{(i)}\beta - y^{(i)}) x_j^{(i)} + \lambda \beta_j^{(*)}) \quad \text{for each } j = 1 \text{ to } n$$

}

$$\beta_j^{(t+1)} = \beta_j^{(t)} - \frac{\alpha}{m} \sum_{i=1}^{m} (x^{(i)}\beta - y^{(i)}) x_j^{(i)} - \frac{\alpha\lambda}{m} \beta_j^{(t)}$$

regularized: $\beta_j^{(t+1)} = (1 - \alpha\lambda)\beta_j^{(t)} - \frac{\alpha}{m} \sum_{i=1}^{m} (x^{(i)}\beta - y^{(i)}) x_j^{(i)}$

unregularized: $\beta_j^{(t+1)} = \beta_j^{(t)} - \frac{\alpha}{m} \sum_{i=1}^{m} (x^{(i)}\beta - y^{(i)}) x_j^{(i)}$

So, difference is that in every epoch our β_j is automatically reduced by factor of $(1 - \alpha\lambda)$

so, automatical $1 \to 0.9999$

$\beta_j \downarrow$ even if $\sum_{i=1}^{m} (x^{(i)}\beta - y^{(i)}) x_j^{(i)}$ zero out

$(1 - \alpha\lambda)$ causes weight decay and act as a multiplicative weight shrinkage at every step

if gradient zero then

$$\beta_j^{(t+1)} = (1 - \alpha\lambda)\beta_j^{(t)}$$

after k steps

$$\beta_j^{(k)} = (1 - \alpha\lambda)^k \beta_j^{(0)}$$

exponential shrink toward 0

$L_2 \to$ weight decay

SGD

repeat {

for $i = 1$ to m {

$$\beta_0 := \beta_0 - \alpha (x^{(i)}\beta - y^{(i)}) x_0^{(i)}$$

$$\beta_j := \beta_j - \alpha((x^{(i)}\beta - y^{(i)}) x_j^{(i)} + \lambda \beta_j) \quad \text{for each } j$$

}

}

$$\beta^{(t+1)} = \beta^{(t)}(1-\alpha\lambda) - \alpha(x^{(i)}\beta - y^{(i)})\, x_j^{(i)}$$

(1−αλ): shrinkage effect

Mini-BGD

[illegible]

A Batch of size b then have m/b such batches:

$$J(\beta) = \frac{1}{2b}\left(\sum_{i=1}^{b}(x_i\beta - y_i)^2 + \lambda\sum_{j=1}^{n}\beta_j^2\right)$$

repeat {
  For $i = 1$ to B {
    $\beta_0 := \beta_0 - \frac{\alpha}{b}\left(\sum_{t=1}^{b}(x_t^{(i)}\beta - y_t^{(i)})\, x_0^{(i)(t)}\right)$ [illegible]
    $\beta_j := \beta_j - \frac{\alpha}{b}\left(\sum_{t=1}^{b}(x_t^{(i)}\beta - y_t^{(i)})\, x_j^{(i)(t)} + \lambda\beta_j\right)$
  }
}

$$\beta^{(t+1)} = \beta^{(t)}(1-\alpha\lambda) - \frac{\alpha}{b}\sum_{t=1}^{b}(x_t^{(i)}\beta - y_t^{(i)})\, x_j^{(i)(t)}$$

(1−αλ): shrinkage; $\frac{\alpha}{b}\sum_{t=1}^{b}$: for a Batch $B_{(i)}$

New weight = (1−αλ) × old weight − α × Data Gradient

(1−αλ) × old weight: Decay / shrinkage, pull β toward 0

α × Data Gradient: fit data, fitting data

at Ridge Optimal ω^*

$$\nabla J(\omega^*) = \nabla MSE(\omega^*) + \lambda\omega^* = 0$$
$$\nabla MSE(\omega^*) = -\lambda\omega^*$$

So, gradient descent is not zero at Ridge optimal (unless $\omega^* = 0$)

$$\text{data gradient} = -\lambda\omega^*$$

exactly balances shrinkage

So, $\omega^{(t+1)} = \omega^{(t)}(1-\alpha\lambda) - \alpha(-\lambda\omega^*)$

$\omega^{(t+1)} = \omega^*$

Once reach optimal Ridge, the decay is continuously canceled by data gradient

So, that why weights don't keep shrinking forever.

In d2 (Ridge) Regression

BGD: repeat { $\beta_0 := \beta_0 - \frac{\alpha}{m}\sum_{i=1}^{m}(x^{(i)}\beta - y^{(i)})\, x_0^{(i)}$

$\beta_j := \beta_j - \frac{\alpha}{m}\sum_{i=1}^{m}\left((x^{(i)}\beta - y^{(i)})\, x_j^{(i)} + \lambda\beta_j\right)$ for each $j = 1$ to n

$$\beta_j^{(t+1)} = \beta_j^{(t)}(1-\alpha\lambda) - \frac{\alpha}{m}\sum_{i=1}^{m}(x^{(i)}\beta - y^{(i)})\, x_j^{(i)}$$

$(1-\alpha\lambda)$: weight decay; $\frac{\alpha}{m}\sum_{i=1}^{m}$: fit data

SGD: repeat { [illegible]

For $i = 1$ to m {

$\beta_0 := \beta_0 - \alpha(x^{(i)}\beta - y^{(i)})\, x_0^{(i)}$

$\beta_j := \beta_j - \alpha\left((x^{(i)}\beta - y^{(i)})\, x_j^{(i)} + \lambda\beta_j\right)$ for each $j = 1$ to n

}

}

$$\beta_j^{(t+1)} = \beta_j^{(t)}(1-\alpha\lambda) - \alpha(x^{(i)}\beta - y^{(i)})\, x_j^{(i)}$$

Mini BGD: for each batch size b total $\beta = m/b$

repeat {

for i = 1 to β {

$\beta_0 := \beta_0 - \frac{\alpha}{b}\left(\sum_{k=1}^{b}(x^{(i)(k)}\beta - y^{(i)(k)})x_0^{(i)(k)}\right)$

$\beta_j := \beta_j - \frac{\alpha}{b}\left(\sum_{k=1}^{b}((x^{(i)(k)}\beta - y^{(i)(k)})x_j^{(i)(k)} + \lambda\beta_j)\right)$

}

}

$$\beta^{(t+1)} = \beta^{(t)}(1-\alpha\lambda) - \frac{\alpha}{b}\sum_{k=1}^{b}(x^{(i)(k)}\beta - y^{(i)(k)})x_j^{(i)(k)}$$

at each step our $\nabla J(\beta)$ don't become 0 instead at Optimal β^* it becomes 0 & we know when to abort $\nabla J(\beta^*) = \lambda\beta^* + \nabla MSE(\beta^*) = 0$

$$\nabla MSE(\beta^*) = -\lambda\beta^*$$

$$\nabla MSE = \beta^* - \frac{\alpha}{m}\Sigma(x\beta - y^{(i)})x_j^{(i)}$$

so, at β^* $\beta^{(t)} = \beta^*$

$$\beta^{(t+1)} = \beta^{(t)}(1-\alpha\lambda) + \alpha\lambda\beta^* = \beta^*$$

$$\beta^{(k)} = (1-\alpha\lambda)^k\beta^{(0)}$$

→ multiplicative shrinkage

1.1.2 Spectral Shrinkage & condition Number Stabilization in Ridge: how ridge scales principal components of design matrix in SVD basis & derive regularized condition number (K)

$$J(\beta^R) = \frac{1}{2m}\left(\sum_{i=1}^{m}(x^{(i)}\beta - y^{(i)})^2 + \lambda\sum_{j=1}^{n}\beta_j^2\right)$$

$$= \frac{1}{2m}\left((X\beta - Y)^T(X\beta - Y) + \lambda\beta^T\Lambda\beta\right)$$

$\Lambda = diag(0, 1, 1, \ldots, 1)$

$$J(\beta) = \frac{1}{2m}\left(Y^TY - 2Y^TX\beta + \beta^TX^TX\beta + \lambda\beta^T\Lambda\beta\right)$$

$$\nabla J(\beta) = \frac{1}{2m}\left(X^T(X\beta - Y) + \lambda\Lambda\beta\right)$$

$$\nabla^2 J(\beta) = \frac{1}{m}\left(X^TX + \lambda\Lambda\right) = H$$

$\frac{X^TX}{m} \geq 0$, $\frac{\lambda\Lambda}{m} > 0$ } > 0

so, absolutely greater zero

also, $\Lambda = I$

$H = \frac{1}{m}(X^TX + \lambda I)$ } so, global local always (up bowl)

$X^TX = QDQ^T$ $\quad D \to diag(e_1, e_2, \ldots e_d)$ e → eigen

$QQ^T = Q^TQ = I$

$$H = \frac{1}{m}(QDQ^T + Q(\lambda I)Q^T)$$

$$= \frac{1}{m}Q(D + \lambda I)Q^T$$

so, $(D + \lambda I) \to diag(e_1 + \lambda, e_2 + \lambda, \ldots, e_d + \lambda)$

so, our principal axes remain same just the eigenvalue of H which become added to λ $(\lambda > 0)$. So, H is always upward bowl.

$$K = \frac{\lambda_{max} + \lambda}{\lambda_{min} + \lambda}$$

$X\Pi \to \lambda$ eigenvalue

$\lambda_i \to Q_i^TX^TX$

as well if $X = U\Sigma V^T$

unregularized $\frac{X^TX}{m} = \text{Hessian} = \frac{V\Sigma^T\Sigma V^T}{m}$

$\Sigma = \text{diag}(u_1, u_2, \ldots, u_d)_{m \times d}$

$\Sigma^T\Sigma = \text{diag}(u_1^2, u_2^2, \ldots, u_d^2)_{d \times d}$, or

$H = \frac{1}{m}(X^TX + \lambda I) = \frac{1}{m}(V\Sigma^2V^T + V(\lambda I)V^T)$

$H = V\left(\frac{\Sigma^2 + \lambda I}{m}\right)V^T$

$\hat{\beta}_{ridge} = (X^TX + \lambda I)^{-1}X^TY$

$= V(\Sigma^2 + \lambda I)^{-1}\Sigma^TU^TY$

$(\Sigma^T\Sigma + \lambda I)^{-1} = \text{diag}\left(\frac{1}{u_1^2+\lambda}, \frac{1}{u_2^2+\lambda}, \ldots, \frac{1}{u_d^2+\lambda}\right)$

$\hat{\beta}_{ridge} = V\,\text{diag}\left(\frac{u_i}{u_i^2+\lambda}\right)U^TY$

$\hat{\beta}_{OLS} = (X^TX)^{-1}X^TY = V\,\text{diag}\left(\frac{1}{u_i}\right)U^TY$

$\frac{u_i}{u_i^2+\lambda} = \frac{1}{u_i}\left(\frac{u_i^2}{u_i^2+\lambda}\right)$

$\hat{\beta}_{ridge} = V\,\text{diag}\left(\frac{u_i^2}{u_i^2+\lambda}\right)\text{diag}\left(\frac{1}{u_i}\right)U^TY$

$= V\,\text{diag}\left(\frac{u_i^2}{u_i^2+\lambda}\right)V^T\hat{\beta}_{OLS}$

shrinkage factor

each principal component v_i scaled $S_i = \frac{u_i^2}{u_i^2+\lambda}$

so, $0 < S_i < 1$

* for large u_i

$u_i^2 \gg \lambda \Rightarrow S_i \approx 1$ so strong dirⁿ are barely changed

* for small u_i

$u_i^2 \ll \lambda$

$S_i \approx \frac{u_i^2}{\lambda} \ll 1$

unstable dirⁿ are heavily damped

$u_i \rightarrow$ eigenvalue of X

so, eigenvalues of H are $(u_i^2 + \lambda)$

so, $\kappa(H) = \frac{u_{max}^2 + \lambda}{u_{min}^2 + \lambda}$

if $u_{min}^2 \approx 0$

then OLS	L2
$\kappa(H) = \frac{u_{max}^2}{u_{min}^2} \approx \infty$	$\kappa(H) = \frac{u_{max}^2 + \lambda}{\lambda} < \infty$

so, every nearly singular matrix turns well

In L2: $H = \nabla^2 J(\beta) = \frac{1}{m}(X^TX + \lambda\Lambda)$

$v^THv > 0$, always positive definite $\Lambda \approx I$ can be used

$X = U\Sigma V^T$ $\Sigma = \text{diag}(\sigma_1, \sigma_2, \ldots, \sigma_d)_{m \times d}$

$\hat{\beta}_{ridge} = V\,\text{diag}\left(\frac{\sigma_i^2}{\sigma_i^2 + \lambda}\right)V^T\hat{\beta}_{OLS}$

eigenvalues of $H \equiv (\sigma_i^2 + \lambda)$

$\kappa(H) = \frac{\sigma_{max}^2 + \lambda}{\sigma_{min}^2 + \lambda}$ which work well even when $\sigma_{min} \rightarrow 0$

each principal component v_i scaled $S_i = \frac{\sigma_i^2}{\sigma_i^2 + \lambda}$ $0 < S_i < 1$

for large $\sigma_i \Rightarrow \sigma_i^2 \gg \lambda$: $S_i \approx 1$

for small $\sigma_i \Rightarrow \sigma_i^2 \ll \lambda$: $S_i \approx \frac{u_i^2}{\lambda} \ll 1$

Date / Page

1.1.2 Bias & variance / covariance Matrix of Ridge estimator:

$Y = X\beta^* + \varepsilon$

$Bias(\hat{\beta}^R) = E[\hat{\beta}^R] - \beta^*$

$\hat{\beta}^R = (X^TX+\lambda I)^{-1}X^TY$

$E[\hat{\beta}^R] = (X^TX+\lambda I)^{-1}X^TX\,\beta^* = W\beta^*$

$Bias(\hat{\beta}^R) = W\beta^* - I\beta^* = (W-I)\beta^*$

$W = (X^TX+\lambda I)^{-1}X^TX$

$X = U\Sigma V^T,\ X^T = V\Sigma^T U^T \Rightarrow X^TX = V\Sigma^T\Sigma V^T$

$(V(\Sigma^T\Sigma+\lambda I)V^T)^{-1}V\Sigma^T\Sigma V^T$

$W = V(\Sigma^T\Sigma+\lambda I)^{-1}\Sigma^T\Sigma V^T$

$I = V\ I\ V^T$

$W - I \Rightarrow (\Sigma^T\Sigma+\lambda I)^{-1}\Sigma^T\Sigma - I$

$\Sigma^T\Sigma = diag(\sigma_1^2 \cdots \sigma_d^2)$

$\frac{\sigma_i^2}{\sigma_i^2+\lambda} - 1 = \frac{-\lambda}{\sigma_i^2+\lambda}$

$Cov(\hat{\beta}^R) = Var(\hat{\beta}^R) = Var(AY) = A\,Var(Y)A^T = \sigma^2 AA^T$

$Var(Y) = Var(\varepsilon) = \sigma^2 I\ ,\ A = (X^TX+\lambda I)^{-1}X^T$

$A^T = X(X^TX+\lambda I)^{-1}$

$AA^T = (X^TX+\lambda I)^{-1}X^TX(X^TX+\lambda I)^{-T}$

$= (X^TX+\lambda I)^{-1}X^TX(X^TX+\lambda I)^{-1}$

$Cov(\hat{\beta}^R) = \sigma^2(X^TX+\lambda I)^{-1}X^TX(X^TX+\lambda I)^{-1}$

$\Rightarrow \frac{\sigma^2 u_i}{(u_i+\lambda)^2}$

$X^TX \Rightarrow$ eigenvalues $u_1, u_2 \cdots, u_d$

Date / Page

squared-bias $\Rightarrow \left(\frac{\lambda}{u_i+\lambda}\right)^2 \Rightarrow \frac{\lambda^2}{(u_i+\lambda)^2}$

variance $\Rightarrow \sigma^2\left(\frac{u_i}{u_i+\lambda}\right)^2 = \frac{\sigma^2 u_i^2}{(u_i+\lambda)^2}$

initial plane OLS — small $\lambda > 0$ penalty → $\lambda \to \infty$ $\beta = 0$ forced

- Bias = 0
- variance = $\frac{\sigma^2}{u_i}$ = max

- ↑ Bias $\Rightarrow \frac{\lambda^2}{(u_i+\lambda)^2} \Rightarrow$ ↑
- variance $= \frac{\sigma^2 u_i}{(u_i+\lambda)^2}$ ↓

variance − square bias

$\sigma^2 u_i - \lambda^2 > 0$

large — decreases >0 — divide denominator ↑ so, variance faster

$var(\varepsilon_i) = \sigma^2$

Another Bias $(\hat{\beta}^R)$: $X^TX = (X^TX+\lambda I) - \lambda I$

$Bias = \underbrace{(X^TX+\lambda I)^{-1}X^TX}_{I - \lambda(X^TX+\lambda I)^{-1}} - I)\beta$

$Bias(\hat{\beta}^R) = -\lambda(X^TX+\lambda I)^{-1}\beta^*$

$E[\hat{\beta}^R] = (X^TX+\lambda I)^{-1}X^TX\beta$

$Bias(\hat{\beta}^R) = -\lambda(X^TX+\lambda I)^{-1}\beta \Rightarrow \frac{-\lambda}{\lambda+u_i}$

$Cov(\hat{\beta}^R) = \sigma^2(X^TX+\lambda I)^{-1}X^TX(X^TX+\lambda I)^{-1}$

$= \sigma^2 Q\,diag\left(\frac{u_i}{(u_i+\lambda)^2}\right)Q^T \Rightarrow \frac{\sigma^2 u_i}{(u_i+\lambda)^2}$

Ridge variance decreases faster than square bias

$X^TX = Q\Phi Q^T\ ;\ \Phi = diag(u_1, u_2 \cdots, u_d)$

L1L2 ⑤ The Bias-variance trade off in Ridge:

$\text{variance}(\hat{\beta}^R) = \text{Cov}(\hat{\beta}^R) = \sigma^2 (X^TX+\lambda I)^{-1} X^T X (X^TX+\lambda I)^{-1}$

$\text{Bias}(\hat{\beta}^R) = -\lambda (X^TX+\lambda I)^{-1} \beta^*$

$X = U\Sigma V^T$; $V^TV = I$, $U^TU = I$

$X^TX = V\Sigma^T\Sigma V^T \Rightarrow X^TX = V D V^T$ $\quad D = \text{diag}(u_1, u_2 \cdots, u_d)$

$\text{Var}(\hat{\beta}^R) = \sigma^2 (V(D+\lambda I)V^T)^{-1} V D V^T (V(D+\lambda I)V^T)^{-1}$

$\sigma^2 V (D+\lambda I)^{-1} V^{-1} V D V^T V^{-T} (D+\lambda I)^{-1} V^T$

$= \sigma^2 V \, \text{diag}\left(\frac{u_i}{(u_i+\lambda)^2}\right) V^T$

$\text{Bias}(\hat{\beta}^R) = -V \, \text{diag}\left(\frac{\lambda}{u_i+\lambda}\right) V^T \beta^*$

$V_a(\lambda) = \text{total parameter variance} = \text{Tr}(\text{cov}(\hat{\beta}^a)) = \sigma^2 \sum_{i=1}^{d} \frac{u_i}{(u_i+\lambda)^2}$

rate of reduction $(V_a'(\lambda))$

$V_a'(\lambda) = \sigma^2 \sum \frac{-2u_i}{(u_i+\lambda)^3}$

$V_a'(0) = \sigma^2 \sum \frac{-2}{u_i^2} = -2\sigma^2 \sum \frac{1}{u_i^2}$ } -ve

so, variance of parameter ↓

square Bias $= \beta(\lambda) = \|\text{Bias}\|^2 = \sum \beta_i^2 \left(\frac{\lambda}{u_i+\lambda}\right)^2$

for tiny λ : $\frac{\lambda}{\lambda+u_i} \approx \frac{\lambda}{u_i}$ $\Rightarrow \beta(\lambda) = \lambda^2 \sum \frac{\beta_i^2}{u_i^2}$

square Bias $\propto \lambda^2$

rate : 2λ so, grows slowly near 0

Bias-variance Trade Off in Ridge:

the variance rate of decrease fast as ($\lambda > 0$ small)

where as bias rate of increase is slow ($\lambda > 0$ small)

so, initially $\lambda > 0$; decrease in total parameter variance faster then squared bias increase

as rate of variance $\propto \left(-2\sigma^2 \sum \frac{u_i}{(u_i+\lambda)^3}\right)$ decrease rate

rate of squared bias $\propto (2\lambda)$ increase rate

L1L2 ⑥ Lasso (L_1) Regression Objective

$J(\beta^L) = \frac{1}{2m}\left(\sum_{i=1}^{m} (x^{(i)}\beta - y^{(i)})^2 + \lambda \sum_{j=1}^{n} |\beta_j|\right)$

$\sum_{j=1}^{n} |\beta_j| \leq S$

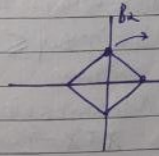
$|\beta_j|$ or $|a| \xrightarrow{+ve} +a$, $\xrightarrow{-ve} -a$

$\beta_{OLS} = (X^TX)^{-1}X^TY$

$J(\beta) = \frac{1}{2m}\left((X\beta - Y)^T(X\beta - Y) + \lambda \sum_{j=1}^{n} |\beta_j|\right)$ $\quad \lambda \approx I$

$(\beta^TX^T - Y^T)(X\beta - Y) = \beta^TX^TX\beta - 2Y^TX\beta + Y^TY$

$j=2$ $\quad |\beta_1| + |\beta_2| \leq C$

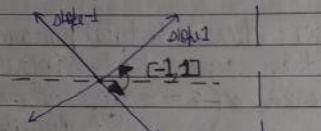


sharp corner at axis

non-differentiable at corner point

but it also gives us the chance for β_j to become zero

$$\partial|x| \begin{cases} 1 & x>0 \\ [-1,1] & x=0 \\ -1 & x<0 \end{cases}$$



$$\nabla J(\beta_0) = \frac{1}{m}\left(X^T(X\beta - Y) + \lambda \partial|\beta|\right) = 0$$

$$\beta = [\beta_0\ \beta_1\ \cdots\ \beta_n]$$
$$\partial\beta = [\partial\beta_1\ \cdots\ \partial\beta_n]$$

$$\therefore \nabla J(\beta_j) = \frac{1}{m}\sum_i x_j^{(i)}(x^{(i)}\beta - y) + \lambda\,\partial|\beta_j| = 0$$

$$\sum_i x_j^{(i)}(x^{(i)}\beta - y) = -\lambda\,\partial|\beta_j|$$

then $\beta_j = 0$

$$\therefore \left|\frac{1}{m} X_j^T(X\beta - Y)\right| \le \lambda \quad [-\lambda, \lambda]$$

so, b/w λ & $-\lambda$ coefficient can stay exactly 0.

So,
non zero $\beta_j \Rightarrow$ MSE grad $= \pm\lambda$
zero $\beta_j \Rightarrow$ |MSE grad| $\le \lambda$

scaled X

If our col^m of orthogonal in X then $X^TX = I$ orthonormal

$$\hat{\beta}_{OLS} = X^TY$$

$$J(\beta) = \frac{1}{2m}\left((X\beta - Y)^T(X\beta - Y) + \lambda\sum_{j=1}^{n}|\beta_j|\right)$$

$\beta^T X^T X \beta$ ($X^TX = I$) $\quad -2Y^TX\beta \quad +Y^TY \quad -2\beta^TX^TY$

$$J(\beta) = \frac{1}{2m}\left(Y^TY - 2\beta^T\hat{\beta}_{OLS} + \beta^T\beta + \lambda\sum_{j=1}^{n}|\beta_j|\right)$$

$$J(\beta_j) = \frac{1}{2m}\left(y^Ty + b_j^2 - 2b_j\hat{b}_j + \hat{b}_j^2 - \hat{b}_j^2 + \lambda\sum|b_j|\right)$$

$(b_j - \hat{b}_j)^2$

constant $+\frac{1}{2m}\left((b_j - \hat{b}_j)^2 + \lambda\sum|b_j|\right)$

$$(\beta - \hat{\beta}_{OLS})^T(\beta - \hat{\beta}_{OLS}) + \lambda|\beta|$$

$(\beta^T - \hat{\beta}^T)(\beta - \hat{\beta}) = \beta^T\beta - \beta^T\hat{\beta} - \hat{\beta}^T\beta + \hat{\beta}^T\hat{\beta}$

① $\hat{\beta}_j > 0 \Rightarrow$ $\quad 2\beta - 2\hat{\beta} + \lambda = 0$

$\beta_j = \hat{\beta}_{OLS_j} - \lambda = \hat{\beta}_{R_j}$

$(\hat{\beta}_{OLS_j} > \lambda > 0)$

② $\hat{\beta}_j < 0$ $\quad \hat{\beta}_{R_j} = \hat{\beta}_{OLS_j} + \lambda < 0$

$(\hat{\beta}_{OLS_j} < -\lambda < 0)$

③ $|\hat{\beta}_{OLS_j}| \le \lambda$ $\quad \hat{\beta}_{R_j} = \hat{\beta}_{OLS_j} - \lambda[-1,1] < \lambda \Rightarrow 0$

$\hat{\beta}_{R_j} = 0$

$$\hat{\beta}_{j\,Lasso} = sign(\hat{\beta}_{j\,OLS})\max(|\hat{\beta}_{j\,OLS}| - \lambda, 0)$$

soft thresholding

In Ridge if X^TX can be I $\quad \hat{\beta}_{OLS} = X^TY$

$$\hat{\beta}_R = (X^TX + \lambda I)^{-1}X^TY = \frac{\hat{\beta}_{OLS}}{1+\lambda}$$

$$\hat{\beta}_{j\,Ridge} = \frac{\hat{\beta}_{OLS}}{1+\lambda}$$

So, in lasso when $|\hat{\beta}_{OLS\,j}| \le \lambda$ it means it get to zero then stay there as $|grad\,MSE| \le \lambda$

Lasso Regression (L_1):

$$J(\beta) = \frac{1}{2m}\left(\sum_{i=1}^{m}(x^{(i)}\beta - y^{(i)})^2 + \lambda\sum_{j=1}^{n}|\beta_j|\right)$$

Our $\Sigma|\beta_j|$ is not a differentiable constraint surface but due to this region shape allow $\beta_j = 0$ even when $\hat{\beta}_{OLS} \neq 0$

$$\partial|x| \to \begin{cases} 1 & x>0 \\ [-1,1] & x=0 \\ -1 & x<0 \end{cases}$$

Allows our β_j to reduced to zero when $|\hat{\beta}_j^{OLS}| < \lambda$

When our data is orthonormal such $X^TX = I$

$$\hat{\beta}_j^{Lasso} = \text{sign}(\hat{\beta}_j^{OLS})\max(0, |\beta_j^{OLS}| - \lambda)$$

&

$$\hat{\beta}_j^{Ridge} = \frac{\hat{\beta}_j^{OLS}}{1+\lambda}$$

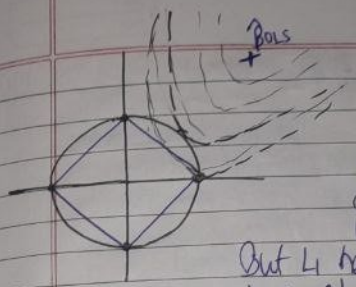
So, through this we can see even when $\hat{\beta}_j^{OLS} \neq 0$ Our $\hat{\beta}_j^{Lasso}$ can yield zero when $|\beta_j^{OLS}| < \lambda$ but our $\hat{\beta}_j^{Ridge}$ can't do such only push them near atmost not reduce to exact zero.

L_1 & L_2 ⇒ Geometric distribution / shape in L_1 for L_1 & L_2 :

If our constraint regions include the $\hat{\beta}_{OLS}$ point then our

$$\hat{\beta}_{OLS} = \hat{\beta}_{Ridge} = \hat{\beta}_{Lasso}$$

but if our $\hat{\beta}_{OLS}$ remain out of region then



in our L_2 due to shape taken to meet the constraint a axes due to curved shape with contours

But L_1 has sharp edges so, its able to melt the region at axes hence reducing them to zero and hence creating sparsity.

$L_1 \Rightarrow$ drop the β_j → creating sharse / parameters

$L_2 \Rightarrow$ retain all β_j → near to zero / not exact

$X \cdot \Rightarrow \hat{\beta}_R = (X^TX + \lambda I)^{-1}X^TY$

$X_{new} = SX$

$\beta_{R_{new}} = (X^TS^TSX + \lambda I)^{-1}X^TSY$

can't be reduced interms of old β_R

So, clearly $Y = X\beta_R$ also changes

So, instead standard normalize each colum before hand best

after centering → $\bar{x}_{ij} = \dfrac{x_{ij}}{\sqrt{\frac{1}{m}\sum_{i=1}^{m}(x_{ij} - \bar{x}_j)^2}}$ } ← for colm

before apply ridge regression its good to centralize and normalize each element w.r.t to its colm.

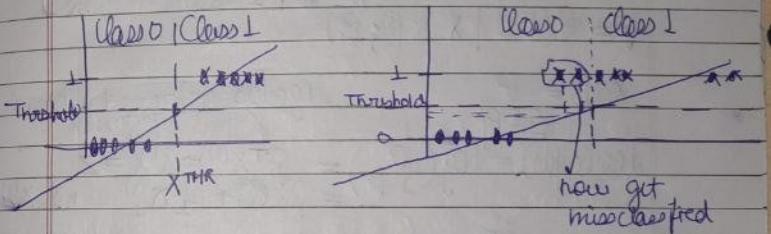
① $\bar{x}_{ij} = x_{ij} - \bar{x}_j$ ② $\bar{x}_{ij} = \dfrac{x_{ij}}{\sqrt{\frac{1}{m}\sum_{i=1}^{m}(x_{ij} - \bar{x}_j)^2}}$

$x_{ij} = \bar{x}_{ij}$

LR ① Failure Model of OLS for classification

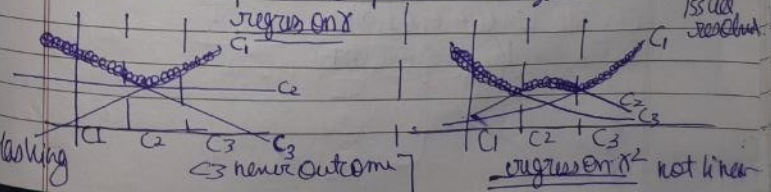
issues on trying to use linear regression on the classification data:

① Outlier extremes disrupt the slope & can cause misclassification of points as slope bends towards our extreme points which cause it to meet our threshold at latter or pt on the x-y divide.



② Linear regression gives us a straight line whose range $(-\infty, \infty)$ but our classify $\{0, 1\}$ with the Threshold 0.5 put on this doesn't look conceptualy interpretable in simple language seems less meaningfull

③ Masking issue for $K \geq 3$. due to major noise in score of some class which may causes it to dominate even in region where it was not supposed to be leading to Output of less then no. of classes present in system.



- If we have K classes then we are needed to regress on X^{K-1} so, not to get Masking issues.

Logistic regression

$y^{(i)} \in \{0, 1\}$ $\quad (x^{(i)}, y^{(i)})_{i=1}^{m}$

parameter $\beta = [\beta_0 \beta_1 \cdots \beta_n]^T$ $\quad x^{(i)} = [1, x_1^{(i)}, x_2^{(i)}, \cdots, x_n^{(i)}]^T$

$p = P(Y=1 \mid X = x^{(i)}; \beta)$

$\text{odds} = \frac{p}{1-p}$ $\quad$ log odds expressed a linear regression

$\log(\text{odds}) = \log\left(\frac{p}{1-p}\right) = \beta^T x^{(i)} = \beta^T x = z$
$= \beta_0 x_0^{(i)} + \beta_1 x_1^{(i)} \cdots + \beta_n x_n^{(i)}$

$\frac{p}{1-p} = e^{\beta^T x} = e^z \Rightarrow p = e^z - e^z p$

$p = \frac{e^z}{1+e^z}$

$h_\beta(x) = p = \frac{e^{\beta^T x}}{1+e^{\beta^T x}} = \frac{1}{1+e^{-\beta^T x}}$

$\sigma(z) = p = \frac{1}{1+e^{-z}}$

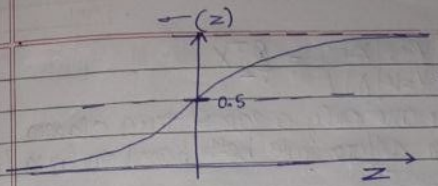
$\sigma(z) \rightarrow$ sigmoid fun

$z = \beta^T x \Rightarrow z \in (-\infty, \infty)$

$e^z \in (0, \infty)$

$1 + e^{-z} \in (1, \infty)$

$1/(1+e^{-z}) \in (0, 1)$



sigmoid fun

Threshold = 0.5

Asymptotes: $y=1$, $y=0$

So, $\sigma(z) = p = 0.5$ (Threshold)

$\text{odds} = \frac{p}{1-p} = 1 \Rightarrow \log \text{odds} = 0 = \beta^T x$

So, Threshold Boundary: $\beta^T x = 0$

Class 1 ⇒	Class 0 ⇒
$p \geq 0.5$	$p < 0.5$
$\frac{1}{1+e^{-z}} \geq 0.5$	$\frac{1}{1+e^{-z}} < 0.5$
$2 \geq 1 + e^{-z} \Rightarrow 1 \geq e^{-z} \Rightarrow 0 \geq -z$	$2 < 1 + e^{-z}$ $\quad 1 < e^{-z}$
$z \geq 0$	$z < 0$
$\beta^T x \geq 0$	$\beta^T x < 0$
Classify as 1 class	Classify as 0 class

Work only in Binary classification

But we won't only have two classes but have K classes then what:

① like suppose we take / find the Prob of a class w.r.t a fixed anchor class then compare all to find which class it fit ⇒

we anchor our K^{th} class

and know: $\sum_{R=1}^{K} P(Y=k \mid X; \beta_k) = 1$

$$\log\left(\frac{P(Y=1|X)}{P(Y=K|X)}\right) = \beta_1^T X$$

It means if we are only a pair of two classes then we can differentiate both based on linear line

so, $\log\left(\frac{P(Y=2|X;\beta_2)}{P(Y=K|X)}\right) = \beta_2^T X$

$$\log\left(\frac{P(Y=k|X;\beta_k)}{P(Y=K|X)}\right) = \beta_k^T X$$

$$P(Y=k|X;\beta_k) = P(Y=K|X) \times e^{\beta_k^T X}$$

known property $\sum_{k=1}^{K} P(Y=k|X;\beta_k) = 1$

$$\sum_{k=1}^{K-1} P(Y=k|X;\beta_k) + P(Y=K|X) = 1$$

$$P(Y=K|X)\left(\sum_{k=1}^{K-1} e^{\beta_k^T X}\right) + P(Y=K|X) = 1$$

$$P(Y=K|X) = \frac{1}{1+\sum_{k=1}^{K-1} e^{\beta_k^T X}}$$

$$P(Y=k|X;\beta_k) = \frac{e^{\beta_k^T X}}{1+\sum_{i=1}^{K-1} \exp(\beta_i^T X)}$$

so we have

$$P(y=k|X) \begin{cases} \frac{1}{1+\sum_{i=1}^{K-1} \exp(\beta_i^T X)} & ; k=K \\ \frac{\exp(\beta_k^T X)}{1+\sum_{i=1}^{K-1} \exp(\beta_i^T X)} & ; k \in [1, K-1] \end{cases}$$

then $\underset{k}{\text{argmax}}\ P(y=k|X)$ find it get k that will be our class for given input of $x^{(i)}$

= although this method works well but we have take a pivot or say anchor which makes it dependent somewhat as well as asymmetric as well depend on which class we take as anchor in start

enclosed version

② soft max Method for K classes

$$P(Y=k|X) = \frac{e^{\beta_k^T X}}{\sum_{i=1}^{K} e^{\beta_i^T X}}$$

so, now $\sum_{k=1}^{K} P(Y=k|X) = \frac{\sum_{k=1}^{K} \exp(\beta_k^T X)}{\sum_{i=1}^{K} \exp(\beta_i^T X)} = 1$ that should be

• so, works symmetric & no anchor to other class

$$\underset{k}{\text{argmax}}\ P(y=k|X)$$

$h_\beta(x) = p \Rightarrow P(y^{(i)} | x^{(i)}; \beta) = (h_\beta(x))^{y^{(i)}} (1 - h_\beta(x))^{(1-y^{(i)})}$

$h_\beta(x) = p = \sigma(z) \in$

$\sigma(z) = \frac{1}{1+e^{-z}}$, $1 - \sigma(z) = \frac{e^{-z}}{1+e^{-z}}$

$\sigma'(z) = \frac{+e^{-z}}{(1+e^{-z})^2} = \sigma(z)(1-\sigma(z))$

$\boxed{\sigma'(z) = \sigma(z)(1-\sigma(z))}$

$z = \beta^T x = \beta_0 + x_1\beta_1 + \beta_2 x_2 \cdots + \beta_n x_n$

$X = [1\ x_1 \cdots x_n]$

Lets look a likelihood → parameter → $\theta \rightarrow \beta$

$L(\theta) = P(\vec{y} | X; \theta) = \prod_{i=1}^{m} P(y^{(i)} | x^{(i)}; \beta)$

$\log L(\theta) = \ell(\theta) = \sum_{i=1}^{m} \left(y^{(i)} \log(\sigma(z)) + (1-y^{(i)}) \log(1-\sigma(z)) \right)$

MLE :

$\frac{\partial \ell(\theta)}{\partial \theta_j} = \sum_{i=1}^{m} \left(\frac{y^{(i)}}{\sigma(z)} \sigma(z)(1-\sigma(z)) x_j - \frac{(1-y^{(i)})}{1-\sigma(z)} \sigma(z)(1-\sigma(z)) x_j \right)$

$\frac{\partial \ell(\theta)}{\partial \theta_j} = \sum_{i=1}^{m} \left(y^{(i)} x_j^{(i)} - \sigma(z) y^{(i)} x_j^{(i)} - \sigma(z) x_j^{(i)} + \sigma(z) y^{(i)} x_j^{(i)} \right)$

$= \sum_{i=1}^{m} \left(y^{(i)} - \sigma(z^{(i)}) \right) x_j^{(i)}$

$\frac{\partial \ell(\theta)}{\partial \theta} = X^T (Y - h_\beta(x))$ →

$\frac{\partial^2 \ell(\theta)}{\partial \theta^2} = -X^T h_\beta(x)(1 - h_\beta(x)) X$ } - we DO Maximum DO, MLE

DO, we (=no curvature) want steepest descent

$\theta_j := \theta_j + \alpha \frac{\partial \ell(\theta)}{\partial (\theta_j)}$

$\theta_j := \theta_j + \alpha \sum (y^{(i)} - \sigma(z^{(i)})) x_j^{(i)}$

$\theta_j := \theta_j - \alpha \sum_{i=1}^{m} (h_\theta(x) - y^{(i)}) x_j^{(i)}$

$X = [1\ X_1\ X_2 \cdots X_n]$ for m=1 to m Samples

$\boxed{\beta := \beta - \alpha X^T (h_\beta(x) - Y)}$

$\frac{\partial \ell(\theta)}{\partial \theta_j} = \sum_{i=1}^{m} (y^{(i)} - \sigma(z^{(i)})) x_j^{(i)}$

$\frac{\partial^2 \ell(\theta)}{\partial \theta_j \partial \theta_k} = \sum_{i=1}^{m} -\sigma(z^{(i)})(1-\sigma(z^{(i)})) x_j^{(i)} x_k^{(i)}$

{ from 1 to n }

DOW see the update method

$\beta := \beta - \alpha X^T (h_\beta(x) - Y)$

is DOW step by step method that take many iterations

DO to short this lets use

Newton raphson Method:

$\theta^{(t+1)} = \theta^{(t)} - \frac{\ell'(\theta)}{\ell''(\theta)} = \theta^{(t)} - (\ell''(\theta))^{-1} \ell'(\theta)$

Date / Page

$$\beta^{(t+1)} = \beta^{(t)} - \left(\frac{\partial^2 l}{\partial\beta\partial\beta^T}\right)^{-1} \frac{\partial l}{\partial\beta}$$

$$\nabla l = \frac{\partial l}{\partial\beta} = X^T(Y - h_\beta(X)) = X^T(Y-P) \qquad P^T X$$

$$\nabla^2 l = H = \frac{\partial^2 l}{\partial\beta\partial\beta^T} = -X^T(h_\beta(X))(1-h_\beta(X))X = -X^T W X$$

let W = square symm. diag $(P^{(i)}(1-P^{(i)}))$

$P = [P_1, \dots, P_m]^T$, $(1-P) = [(1-P_1), \dots (1-P_m)]^T$

$$\beta^{(t+1)} = \beta^{(t)} + (+X^T W X)^{-1} X^T (Y-P)$$

$(X^T W X)^{-1} X^T W X$

$$\beta^{(t+1)} = (X^T W X)^{-1} X^T (WX\beta^{(t)} + (Y-P))$$
$$= (X^T W X)^{-1} X^T W (X\beta^{(t)} + W^{-1}(Y-P))$$
$$= (X^T W X)^{-1} X^T W Z$$

$$Z = (X\beta^{(t)} + W^{-1}(Y-P)) = (X\beta^{(t)} - W^{-1}(Y - h_\beta(X)))$$

$\operatorname{argmin} (Z - X\beta)^T W (Z - X\beta)$

iterated reweight least square (IRLS)

$$\beta^{(t+1)} = \beta^{(t)} - (H)^{-1} \nabla_\beta l$$

$$H = -X^T W X$$

$W = \text{diag}(P^{(i)}(1-P^{(i)}))$

Date / Page

we know our $J(\beta) = \frac{1}{m} \Sigma$ cost fun^n

cost fun^n $\begin{cases} -\log(h_\beta(x)) \to y^{(i)} = 1 \\ -\log(1-h_\beta(x)) \to y^{(i)} = 0 \end{cases}$ $\quad h_\beta(x) = \sigma(z) = \beta \in (0,1)$



cost fun^n $= -y^{(i)} \log(h_\beta(x^{(i)})) - (1-y^{(i)}) \log(1 - h_\beta(x^{(i)}))$

$(h_\beta(x^{(i)}), y^{(i)})$

$$\Rightarrow J(\beta) = -\frac{1}{m} \sum_{i=1}^{m} \left(y^{(i)} \log(h_\beta(x^{(i)})) + (1-y^{(i)}) \log(1-h_\beta(x^{(i)})) \right)$$

$$\frac{\partial J(\beta)}{\partial \beta_j} = -\frac{1}{m} \sum_{i=1}^{m} \left(\frac{y^{(i)}}{h_\beta(x^{(i)})} \times h_\beta(x^{(i)}) \times (1-h_\beta(x^{(i)})) \, x_j^{(i)} - \frac{(1-y^{(i)})}{1-h_\beta(x^{(i)})} \times h_\beta(x^{(i)}) \times (1-h_\beta(x^{(i)})) \, x_j^{(i)} \right)$$

$$\frac{\partial J(\beta)}{\partial \beta_j} = \frac{1}{m} \sum_{i=1}^{m} \left(y^{(i)} - h_\beta(x^{(i)}) \right) x_j^{(i)}$$

$$\frac{\partial^2 J(\beta)}{\partial \beta_j \partial \beta_k} = +\frac{1}{m} \sum_{i=1}^{m} h_\beta(x^{(i)})(1-h_\beta(x^{(i)})) \, x_k^{(i)} x_j^{(i)}$$

H hessian matrix $\quad H = \partial J / \partial\beta \partial\beta^T$

$$H_{ij} = \frac{\partial^2 J(\beta)}{\partial \beta_j \partial \beta_k} = \frac{1}{m} \sum_{i=1}^{m} x_k^{(i)} h_\beta(x^{(i)})(1-h_\beta(x^{(i)})) \, x_j^{(i)}$$
$$x_k^{(i)} S^{(i)} x_j^{(i)}$$

$$H = \frac{1}{m} X^T W X$$

$W = \text{diag}(S_1, S_2, \dots, S_m)$

$S_i = P_i(1-P_i)$ $\qquad P_i = h_\beta(x^{(i)})$

$$X = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1n} \\ 1 & & & & \\ \vdots & & & & \\ 1 & x_{m1} & \cdots & & x_{mn} \end{bmatrix}$$

d = 1th features
m samples

$X^T = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ x_{11} & x_{21} & \cdots & x_{m1} \\ x_{12} & & & \\ x_{1n} & x_{2n} & \cdots & x_{mn} \end{bmatrix} \in R^{(d+1)m}$

$W = \text{diag}\left(\hat{p}_{(1)}(1-\hat{p}_{(1)}), \ldots, \hat{p}_{(m)}(1-\hat{p}_{(m)})\right)$

$X^T W X = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ x_{11} & x_{21} & \cdots & x_{m1} \\ x_{12} & & & \\ x_{1n} & x_{2n} & \cdots & x_{mn} \end{bmatrix} \begin{bmatrix} s_1 & 0 & \cdots & 0 \\ 0 & s_2 & & \\ \vdots & & \ddots & \\ 0 & & & s_m \end{bmatrix} \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1n} \\ 1 & x_{21} & & & x_{2n} \\ \vdots & & & & \\ 1 & x_{m1} & & & x_{mn} \end{bmatrix}$

$(d+1)\times m \quad m\times m \quad m\times(d+1)$

$= \begin{bmatrix} s_1 & s_2 & \cdots & s_m \\ s_1 x_{11} & s_2 x_{21} & \cdots & \\ s_1 x_{12} & & & \\ s_1 x_{1n} & & & s_m x_{mn} \end{bmatrix} \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1n} \\ 1 & x_{21} & & & x_{2n} \\ \vdots & & & & \\ 1 & x_{m1} & & & x_{mn} \end{bmatrix}$

$Q_{11} = s_1 + s_2 \cdots + s_m$

$Q_{12} = s_1 x_{11} + s_2 x_{21} + s_3 x_{31} \cdots + s_m x_{m1}$

$Q_{1n} = s_1 x_{1n} + s_2 x_{2n} x_{2n} \cdots + s_m x_{m1} s_{mn}$

$Q_{ij} = \sum_{i=1}^{m} x_k^{(i)} x_j^{(i)} s^{(i)} \in$

$Q_{nn} = \left(\sum_{i=1}^{m} x_n^{(i)} x_n^{(i)} s^{(i)}\right) = x_{1n} x_{1n} s_1 + x_{2n} x_{2n} s_2 + \cdots$

$H = \frac{1}{m} X^T W X \rightarrow$ symmetric

diagonizable

$\Sigma \lambda_i \|v_i\|_2^2$

$\lambda_i \rightarrow$ +ve ≥ 0

$s^{(i)} = \underbrace{\hat{p}^{(i)}}_{(0,1)} \underbrace{(1-\hat{p}^{(i)})}_{(0,1)} \rightarrow$ +ve

$S \rightarrow$ diagpositive matrix

So) $H = \frac{1}{m} X^T W X$

$v^T H v = \frac{1}{m} v^T X^T W X v = \frac{1}{m} (Xv)^T W (Xv)$

$W \rightarrow \text{diag}(w_i) \quad i \rightarrow d \Rightarrow w_i \rightarrow$ +ve

$\frac{1}{m} \sum w_i \|x_i\|_2^2 \longrightarrow \geq 0$

global convex

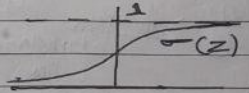
[LR] log-odds assumption & Logistic sigmoid fun

Assumi: log odds to linear for binary classification

$\hat{p} = p(Y=1 \mid X = x^{(i)}; \beta) = h_\beta(x) = \sigma(z)$

$z = \beta^T X \qquad \beta = [\beta_0 \cdots \beta_n]^T, \; X^{(i)} = [1, x_1^{(i)} \cdots x_n^{(i)}]^T$

$\sigma(z) = \frac{1}{1+e^{-z}}$



$\text{odds} = \frac{\hat{p}}{1-\hat{p}}$

$\sigma'(z) = \sigma(z)(1-\sigma(z))$

$z = \beta^T X = \beta_0 + \beta_1 x_1^{(i)} + \cdots + \beta_n x_n^{(i)}$

$\frac{\partial z}{\partial \beta_j} = x_j^{(i)}$

$\sigma(z)$ asymptote lines $y = 0 \; \& \; y = 1$

$\sigma(z) \in (0,1)$

LR ② Decision Boundary ⇐ Probability contour Geometry:

At contour our probability = be P^c

then odds = $\frac{P^c}{1-P^c}$ = Constant

$\beta^T X = C$

so, basically slope will remain same

$\beta^T X = 0$

say $d=3$

$\beta_0 + \beta_1 X_1 + \beta_2 X_2 = 0 = \beta^T X^{(1)}$

say we have two point $(X^{(1)}, X^{(2)})$ on (X_1, X_2) each on Boundary line

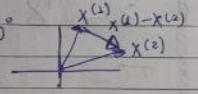
so, $\beta^T X^{(1)} = \beta^T X^{(2)}$

$\beta^T (X^{(2)} - X^{(1)}) = 0$

$\|\beta\|_2 \, \|(X^{(2)} - X^{(1)})\| \cos\theta = 0$

$\theta = 90°$

$X^{(2)} - X^{(1)}$ → decision boundary line



So, β is perpendicular to our decision boundary line.

Similar in contour

$\beta^T X = C$

basically for diff C we see that slope same just intercept changes i.e hyperplanes of counters all parallel planes

Constant probability contours are parallel hyperplanes

$\sigma(z) = 0.5$; $\beta^T X = 0$ is the decision boundary

when $\sigma(z) \geq 0.5$; $\beta^T X \geq 0$ class 1

$\sigma(z) < 0.5$; $\beta^T X < 0$ class 0

β is perpendicular (normal) to the decision boundary line

Constant probability contours are parallel hyperplanes that are ⊥ to the β.

$\beta^T X = \beta^T x + \beta_0 = \log\left(\frac{P}{1-P}\right) = C$

ILR ④ Pathology of Mean squared error for logistic regression Classification: derive the gradient of squared loss applied to a sigmoid hyperplane & prove it causes non-convexity & vanishing gradient on confidently misclassified problems.

Mean square error ⇒ $(h_\beta(x) - Y)^2$ = cost fun

if take $J(\beta) = \frac{1}{2m} \sum (h_\beta(x^{(i)}) - Y^{(i)})^2$

$\sigma'(z) = \sigma(z)(1-\sigma(z))$ $\quad h_\beta(x) = \sigma(z) = \frac{1}{1+e^{-z}}$

$\frac{\partial J(\beta)}{\partial \beta_j} = \frac{2}{2m} \sum (h_\beta(x^{(i)}) - Y^{(i)}) \, h_\beta(x)(1 - h_\beta(x^{(i)})) \times x_j^{(i)}$

$\frac{\partial J(\beta)}{\partial \beta} = \frac{1}{m} X^T W (h_\beta(X) - Y)$

$\frac{\partial J(\beta)}{\partial \beta} = \frac{1}{m} \sum \left(h_\beta(x^{(i)})(1 - h_\beta(x^{(i)})) x_k^{(i)} \, h_\beta(x^{(i)})(1 - h_\beta(x^{(i)})) x_j^{(i)} + (h_\beta(x^{(i)}) - y^{(i)}) \, h_\beta(x^{(i)})(1 - h_\beta(x^{(i)})) x_k^{(i)} \right)$

$\frac{1}{m} \sum \left(\beta_0 \beta_1 x_k^{(i)} x_j^{(i)} + (\beta_0 - y^{(i)}) x_j^{(i)} x_k^{(i)} \right) \beta_1 (1 - \beta_1)$

$\frac{1}{m} \sum \beta_0 \beta_1 (\beta_0 - \beta_1^2 - y^{(i)}) \beta_0 + \beta_1^2 + \beta_0 = y^{(i)} =$

MSE ⇒ $\frac{\partial J}{\partial \beta} = X^T W (h_\beta(x) - y)$

log cost ⇒ $\frac{\partial J}{\partial \beta} = X^T (h_\beta(x) - y)$

So, MSE for contain W

$W_{(i)} = \hat{p}_{(i)}(1-\hat{p}_{(i)})$

Suppose if we have $y=1$, $\hat{p}=0.01$

we know it's a huge misclassify ($\beta<0.5 \rightarrow y=0$)

We would like to update weights

$\beta := \beta - \alpha \frac{\partial J}{\partial \beta}$ ⇒ MSE ⇒ $\frac{\partial J}{\partial \beta} \equiv \hat{p}_{(i)}(1-\hat{p}_{(i)})$ terms

then

$(0.01)(0.99) = 0.0099$ very small

which is very small

a hard to almost negligible improvement

But this is not the issue in log methods

as $y - \hat{y} = 0.99$ stay

as $z \ll 0$ | $\hat{p} = 1/(1+e^{-z})$

$\hat{p} \approx 0$

then $\hat{p}(1-\hat{p}) \approx 0$ |

as $z \gg 0$ |

$\hat{p} \approx 1$

then $\hat{p}(1-\hat{p}) \approx 0$ | MSE gradient $= (\hat{p}-y)\hat{p}(1-\hat{p})$

$\hat{p}(1-\hat{p})$ ouch to zero

as even server misclassified give $\nabla J \rightarrow 0$

which is also known as Vanishing Gradient problem

$\frac{dJ}{dz} = \frac{1}{m}(\sigma(z) - y)\sigma(1-\sigma) = \frac{1}{m}(S-y)S(1-S)$

$S = \sigma(z)$

$\frac{\partial^2 J}{\partial z^2} = \frac{1}{m} S(1-S)\left[S(1-S) + (S-y)(1-2S)\right]$

if $y=1$ ⇒ $\frac{\partial^2 J}{\partial z^2} = \frac{1}{m} S(1-S)\left[S(1-S) + (S-1)(1-2S)\right]$

$= \frac{1}{m} S(1-S)^2(3S-1)$

if $S < 1/3 \Rightarrow$ -ve, $S > 1/3 \Rightarrow$ +ve

So, not global convex for MSE composed with Sigmoid

MSE (squared error) doesn't work for sigmoid as its gradient will have $\hat{p}(1-\hat{p})$ term when we overconfidently mis callsify anything causes to have vanishing gradient $\nabla J \rightarrow 0$ issue. hence causing almost no update.
As well MSE composed of sigmoid fun is not global convex

LOG

Bernoulli Likelihood & Binary cross-entropy cost

Binary cross entropy cost:

$J(\beta) = -\frac{1}{m}\sum_{i=1}^{m} y^{(i)} \log h_\beta(x^{(i)}) + (1-y^{(i)}) \log(1 - h_\beta(x^{(i)}))$

$\nabla J(\beta) = +\frac{1}{m} X^T (h_\beta(x) - y)$

$H = \nabla^2 J(\beta) = \frac{1}{m} X^T W X \Rightarrow W = \text{diag}(\hat{p}_{(i)}(1-\hat{p}_{(i)}))$

global convex

$\hat{p}(1-\hat{p})$ is always

Date / Page /

[LR] ⑥ Why logistic Regression no closed form soln

when $\nabla J = 0 = \frac{1}{m} X^T(h - Y) = 0$

$X^T h = X^T Y$

$h_\beta(x) = \begin{bmatrix} h_1 \\ h_2 \\ \vdots \\ h_m \end{bmatrix}$

$h_i = \frac{1}{1+e^{-z_i}}$

Do, Over all transcedental form and no closed form soln

Logistic regression has no gloval closed form soln & must be solved iteratively. newton or G.D.

[LR] ⑦ BGD, SGD, Mini BGD for logistic Regression both regularized (L2) & unregularized.

$J(\beta) = -\frac{1}{m}\sum \left(y^{(i)} \log(h_\beta(x)) + (1-y^{(i)}) \log(1-h_\beta(x)) \right)$

How unregularized cost fun

$\frac{\partial J(\beta)}{\partial \beta_j} = -\frac{1}{m}\sum (y^{(i)} - h_\beta(x^{(i)})) x_j^{(i)} = \frac{1}{m}\sum (h_\beta(x^{(i)}) - y^{(i)}) x_j^{(i)}$

BGD: repeat {

$\beta_j := \beta_j - \frac{\alpha}{m}\sum (h_\beta(x^{(i)}) - y^{(i)}) x_j^{(i)}$ all j

}

Date / Page /

SGD: repeat {

for all m samples {

$\beta_j := \beta_j - \alpha \sum (h_\beta(x^{(i)}) - y^{(i)}) \cdot x_j$ for all j

}

}

Mini Batch repeat {

for all k = 1 to m/b {

$\beta_j := \beta_j - \frac{\alpha}{b}\sum_i (h_\beta(x^{ki}) - y^{ki}) x_j^{ki}$ all j

}

}

In all three $h_\beta(x) = \beta = \sigma(z) = \frac{1}{1+e^{-z}}$

$z = \beta^T x$

Now if we regularize it so that β is bound z don't shrink to ∞ or $-\infty$ bounds

If $z \to -\infty$ or ∞ then $\beta = 1$ or 0 that's a itself divider

$J(\beta) = -\frac{1}{m}\left(\sum \left(y^{(i)} \log(\beta) + (1-y^{(i)}) \log(1-\beta) \right) + \lambda \sum_{i=1}^{n} \beta_j^2 \right)$

$\frac{\partial J(\beta)}{\partial \beta_j} = \frac{1}{m}\left(\sum_{i=1}^{m} (y^{(i)} - h_\beta(x^{(i)})) x_j^{(i)} + 2\lambda \beta_j \right)$

$= \frac{1}{m}\sum (h_\beta(x^{(i)}) - y^{(i)}) x_j^{(i)} + \lambda \beta_j$

$\frac{\partial J(\beta)}{\partial \beta} = \frac{1}{m} X^T (h_\beta(x) - Y) + \lambda \beta$

BGD: $\beta^{(t+1)} = \beta^{(t)} - \alpha\left[\frac{1}{m} X^T(h_\beta(x) - y) + \lambda\beta^{(t)}\right]$

$\beta^{(t+1)} = \underbrace{(1-\alpha\lambda)}_{<1}\beta^{(t)} - \frac{\alpha}{m} X^T(h_\beta(x) - y)$

weight decay

SGD: $\beta^{(t+1)} = (1-\alpha\lambda)\beta^{(t)} - \alpha(h_i - y_i)x_i$

MiniBatch: $\beta^{(t+1)} = (1-\alpha\lambda)\beta^{b} - \frac{\alpha}{b}(X_\beta^T)(h_\beta(X^\beta) - Y^\beta)$

In BGD: reg: $\beta := \beta(1-\alpha\lambda) - \frac{\alpha}{m} X^T(h_\beta(x) - y)$

unreg: $\beta := \beta - \frac{\alpha}{m} X^T(h_\beta(x) - y)$

SGD: reg: $\beta := (1-\alpha\lambda)\beta - \alpha(h_i - y_i)x_i$

unreg: $\beta := \beta - \alpha(h_i - y_i)x_i$

MiniBatch: reg: $\beta := \beta(1-\alpha\lambda)\beta - \frac{\alpha}{b}(X_\beta^T)(h_\beta(X^\beta) - Y^\beta)$

unreg: $\beta := \beta - \frac{\alpha}{b} X_\beta^T(h_\beta(X^\beta) - Y^\beta)$

reg → L2

L2 Linear separable data Pathology in Logistic regression & how L2 resolves it

$H = \frac{1}{m} X^T W X$

$W_i = \rho_i(1-\rho_i)$

$\rho(1-\rho)$ $\rho \in (0,1)$ so, $\rho(1-\rho) \in (0, \frac{1}{4}]$

$g = \rho^2 - \rho$

$g' = 2\rho - 2\rho = 0 \Rightarrow \rho = \frac{1}{2}$

$g'' = -2 \Rightarrow$ +ve → maxima



$H \geq 0$

but means can be zero not strictly convex

if we want that

$v^T X \Rightarrow \|Xv\| \neq 0$

then

$Xv \neq 0$

if X col^m are all linear independent

Rank(X) = d

then $Xv = 0$ only if $v = 0$

so, $H > 0$ strictly if X is good boy

If we have β_s that clearly separates data

$\beta_s^T x > 0 \rightarrow C1 \rightarrow +\infty$

$\beta_s^T x < 0 \rightarrow C0 \rightarrow -\infty$

now give $\Rightarrow \beta = C\beta_s$ $C > 0$

then $C\beta_s^T x > 0$

so $C \to \infty$; $\beta \to \infty$ so, $+\infty$ $Z \to \infty$

$C\beta_s^T x < 0$

$C \to \infty$; $\beta \to \infty$. so, $-\infty$ to $-\infty$ $Z \to -\infty$

$y_i = 1 \Rightarrow h_i \to 1 \Rightarrow \log h_i \approx 0$

$y_i = 0 \Rightarrow h_i \to 0 \Rightarrow \log(1-h_i) \approx 0$

$J(\beta) \approx 0$

$h_i(1-h_i) \approx 0$

So, $W_i = h_i(1-h_i) \approx 0$

so, $H = \frac{1}{m} X^T W_i X \approx 0$

To reach small α smaller loss Increase $\|\beta\|$ $J(\beta) \to 0$ $\beta \to \infty$

So Curvature vanish

$\nabla J(\beta) = \frac{1}{m} X^T(h_\beta(x) - y) \to 0$

≈ 0 Optimization slows

So, optimization drifts slowly while norm $\|\beta\|$ keeps growing.

By now L2 regularize the

$$J(\beta) = \frac{1}{m}\sum$$

$$\nabla J_\lambda = \nabla J(\beta) = \frac{1}{m} X^T(h_\beta(x) - y) + \lambda\beta$$

$$H_\lambda = \nabla^2 J(\beta) = \frac{1}{m} X^T W X + \lambda I$$

now now if we have $H \to 0$

then $H_\lambda \to \lambda I$

eigenvalues $\approx \lambda$

So, curvature doesn't disappear

So, if $\beta \to \infty$

error cost $\to 0$

regularization term $\lambda\|\beta\|^2 \to \infty$

So, now don't let $J_\lambda \to \infty$

∞ $|\beta| \to \infty$

So, penalized

$$\|\hat{\beta}_M\| < \infty$$

logistic regression : $\nabla J = \frac{1}{m} X^T(h - y)$; $H = \frac{1}{m} X^T W X$

$W = \text{diag}(h_i(1-h_i))$

- globally convex as $X^T W X \geq 0$
- strictly convex if $\text{Rank}\, X = d$
- if data strictly linear separable, the unregularized optimum occurs at $\|\beta\| \to \infty$ with $h_i \to 0$ or 1, $\lambda_k(W) \to 0$ $\quad W \to 0$, $J(\beta) \to 0$

L2 regularize $\Rightarrow H_\lambda = \frac{1}{m} X^T W X + \lambda I$

So, even if $W \to 0$, $H_\lambda \to \lambda I$

$\|\beta\| \to \infty$ then $J(\beta) \to \infty$ not optimized

$\Rightarrow$ Newton-raphson root finding

$X = \sqrt{100}$

$X^2 = 100$

$f(x) = x^2 - 100 = 0$, $f'(x) = 2x$

$x_j := x_j - \frac{f(x)}{f'(x)}$

$x_j := x_j - \frac{(x_j^2 - 100)}{2x_j}$

$x = 20$ start $\Rightarrow 20 - \frac{(20^2 - 100)}{2 \times 20} = 20 - 7.5$

$x = 12.5 \Rightarrow 12.5 - 2.25 = 10.25$

$x = 10.25 \Rightarrow 10.25 - 0.248$

$x = 10.01 \Rightarrow 10.01 - 0.009$

$x = 10.001 \Rightarrow 10.001 - 0.00099$

$x = 10.000000001$

$x \approx 10$

$x = \sqrt{3}$

$x^2 = 3$

$f(x) = x^2 - 3$

$x_j := x_j - \frac{f(x)}{f'(x)} = x_j - \frac{(x^2 - 3)}{2x}$

$x \approx 1.732$

$x_0 = 10 \Rightarrow 10 - 4.85$

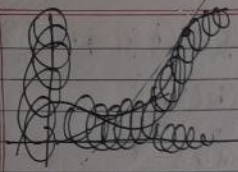
$x_1 = 5.15 \Rightarrow 5.15 - 2.28$

$x_2 = 2.86 \Rightarrow 2.86 - 0.909$

$x_3 = 1.95 \Rightarrow 1.95 - 0.2$

$x_4 = 1.744 \Rightarrow 1.744 - 0.012$

$x_5 = 1.731 \Rightarrow 1.730$



in newton raboon
other
error converges

(A) Cost method

in unregularized our $\nabla J = \frac{1}{m} X^T(h(x) - Y)$

$H = \frac{1}{m} X^T W X$

Unregularized $\nabla J_\lambda = \frac{1}{m} X^T(h(x) - y) + \lambda\beta$

$H_\lambda = \frac{1}{m} X^T W X + \lambda I$

Newton $\beta^{(t+1)} = \beta^{(t)} - (H)^{-1}\nabla J$

unregularized: $\beta^{(t+1)} = \beta^{(t)} - (X^T W X)^{-1} X^T (P - Y)$

$= (X^T W X)^{-1} X^T W X \beta^{(t)} - (X^T W X)^{-1} X^T W W^{-1} P$

$\beta^{(t+1)} = (X^T W X)^{-1} X^T W (X\beta^{(t)} - W^{-1}(P - y))$

$= (X^T W X)^{-1} X^T W z$

similar to LWLR

regularized: $\beta^{(t+1)} = \beta^{(t)} - (X^T W X + \lambda I)^{-1}(X^T(P - Y) + \lambda\beta)$

(B) Log likelihood Maximize

$l(\beta) = \sum [y^{(i)} \log h_i + (1 - y_i) \log(1 - h_i)]$

$\nabla l = X^T(y - h)$

$\nabla^2 l = -X^T W X =$

$\beta_{new} = \beta + (X^T W X)^{-1} X^T (y - h)$

identical to before

$Z = X\beta - W^{-1}(P - y) \Rightarrow Z_i = \underbrace{X_i^T \beta}_{\text{current}} + \underbrace{\frac{y_i - h_i}{h_i(1 - h_i)}}_{\text{newton correction}}$

* Softmax $K = 2$

$P(Y = k|X) = \frac{e^{\beta_k^T X}}{\sum_{k=1}^{K} e^{\beta_k^T X}}$

$P(Y = 0|X) + P(Y = 1|X) = 1$

$P(Y = 1|X) = \frac{e^{\beta_1^T X}}{e^{\beta_0^T X} + e^{\beta_1^T X}}$

$P(Y = 0|X) = 1 - \frac{e^{\beta_1^T X}}{e^{\beta_0^T X} + e^{\beta_1^T X}} = \frac{e^{\beta_0^T X}}{e^{\beta_0^T X} + e^{\beta_1^T X}}$

$P(Y = 1|X) = \frac{1}{1 + e^{(\beta_0 - \beta_1)^T X}}$ $\quad \beta_0 - \beta_1 = \beta$

$= \frac{1}{1 + e^{\beta^T X}} \rightarrow$ Sigmoid fun

Binary softmax = Binary logistic regression

Newton : $\beta_{new} = \beta + (X^T W X)^{-1} X^T (y - h)$

La ⇒ $\beta_{new} = \beta + (X^T W X + \lambda I)^{-1} (X^T (y - h) - \lambda \beta)$

IRLS ⇒ $Z = X\beta + W^{-1}(y - h)$

$\beta_{new} = (X^T W X)^{-1} X^T W Z$

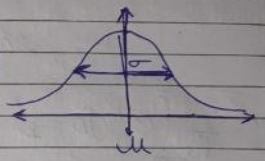
Softmax ⇒ $P(y = k) = \frac{e^{\beta_k^T x}}{\sum_j e^{\beta_j^T x}}$

K=2 ⇒ Softmax = Sigmoid

LDA Univariate Gaussian PDF
Multivariate Gaussian PDF & Contour Geometry

Univariate Gaussian
(One feature)

$X \sim N(\mu, \sigma^2)$



$\mu \rightarrow$ centre
$\sigma \rightarrow$ spread of data

$f_X(x) = C \exp(-a(x-\mu)^2)$

C ?, a ?

① $\int_{-\infty}^{\infty} C \exp(-a(x-\mu)^2) = 1$

② $\sigma^2 = \text{var}(x) = E[(x-\mu)^2] = \int (x-\mu)^2 f_X(x) dx$

$C \int_{-\infty}^{\infty} (x-\mu)^2 e^{-a(x-\mu)^2} dx$

$a(x-\mu)^2 = t$

$(x-\mu)^2 = t/a \Rightarrow (x-\mu) = \sqrt{t/a}$

$2(x-\mu)\partial x = \frac{\partial t}{a}$

$\partial x = \frac{\partial t}{2(x-\mu)a} = \frac{\partial t}{2\sqrt{t}a}$

$C \int_0^{\infty} \sqrt{\frac{t}{a}} e^{-t} \frac{dt}{2\sqrt{t}a}$

$\left(\frac{C}{2a\sqrt{a}}\right) \int_0^{\infty} t^{1/2} e^{-t} dt$

I I

$t = b^2 \Rightarrow b = \sqrt{t}$

$\partial t = 2b\,\partial b$

$\frac{\partial t}{2\sqrt{t}} = \partial b$

$\frac{C}{2a\sqrt{a}} \int_0^{\infty} b e^{-b^2} x$

$-t^{1/2} e^{-t} \Big]_0^{\infty} + \int \frac{e^{-t}}{\sqrt{t}} dt$

$2 \int_0^{\infty} e^{-b^2} db$ } $\sqrt{\pi}$

I

$I = \int_0^{\infty} e^{-x^2} dx$

$I^2 = \frac{1}{2} \int_0^{2\pi} \int_0^{\infty} r e^{-r^2} dr\, d\theta$

$r^2 = t \Rightarrow 2r\,dr = dt$

$-\left[e^{-t}\right]_0^{\infty} = 1$

So, $\frac{C\sqrt{\pi}}{2a\sqrt{a}}$

$I^2 = \pi$

$I = \frac{\sqrt{\pi}}{2}$

$\frac{C\sqrt{\pi}}{2a\sqrt{a}} = \sigma^2$, $C\int_{-\infty}^{\infty} e^{-a(x-u)^2} dx = 1$

$a(x-u)^2 = b^2$

$\sqrt{a}(x-u) = b$

$2a(x-u)\partial x = 2b\partial b$

$\partial x = \frac{b}{a(x-u)}\partial b = \frac{1}{\sqrt{a}}\partial b$

$\frac{C}{\sqrt{a}}\int_{-\infty}^{\infty} e^{-b^2}\partial b = 1$

$\int_0^{2\pi}\int_0^{\infty} re^{-r^2} dr\, d\psi = \frac{1 \times 2\pi}{2} = \pi$

$C = \sqrt{\frac{a}{\pi}}$) $\frac{C\sqrt{\pi}}{2a\sqrt{a}} = \sigma^2$ $a = \frac{1}{2\sigma^2}$

$C = \frac{1}{\sqrt{2\pi}\sigma}$ so

$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(\frac{-(x-u)^2}{2\sigma^2}\right)$

if centred & $\sigma = 1$

$f_Z(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right)$

In Multivariate case

if we $[Z_1, Z_2, \dots, Z_n] = Z$ centered & normalized

$f(z) = f(z_1) \cdots f(z_n) = \prod f(z_i)$

$= \frac{1}{(2\pi)^{n/2}} \exp\left(-\frac{\sum z_i^2}{2}\right) = \frac{1}{(2\pi)^{n/2}} \exp\left(-\frac{z^T z}{2}\right)$

$X = \frac{X+u}{\sigma} \Rightarrow X = u + \sigma Z$

$A^T A = \Sigma$

$|A| = |\Sigma|^{1/2}$

$X = u + AZ$

$Z = A^{-1}(X-u)$

$\frac{\partial Z}{\partial X} = A^{-1} \Rightarrow |J| = |A|^{-1} = \frac{1}{|\Sigma|^{1/2}}$

$\partial Z \to |J|\,\partial X$

$f(x) = \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}} \exp\left(\frac{-1}{2}(X-u)^T A^{-T} A^{-1}(X-u)\right)$

$f(x) = \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(X-u)^T \Sigma^{-1}(X-u)\right)$

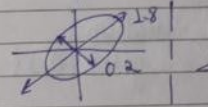
$X \sim N(u, \Sigma)$ ↳ COV

$\Sigma \to$ represents contours shape eigenvalue & eigenvector of it tell the principle axis of contours

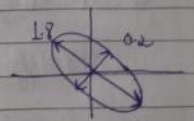
$\Sigma = I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ circle

bump out of 2D plane

$\Sigma = \begin{bmatrix} 1 & 0.8 \\ 0.8 & 1 \end{bmatrix}$



$\Sigma = \begin{bmatrix} 1 & -0.8 \\ -0.8 & 1 \end{bmatrix}$



$\begin{bmatrix} 1-\lambda & 0.8 \\ 0.8 & 1-\lambda \end{bmatrix}$

$(1-\lambda)^2 = 0.8^2$

$1 \pm 0.8 = \lambda$

$1.8, 0.2$

$\begin{bmatrix} 0.8 & 0.8 \\ 0.8 & 0.8 \end{bmatrix}$ $\begin{bmatrix} 0.8 & 0.8 \\ 0.8 & 0.8 \end{bmatrix}$

$\begin{bmatrix} 1 \\ +1 \end{bmatrix}$ $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

(after horizontal stretch) basically gaussian shape

Gaussian / Normal distribution

Univariate: $X \sim N(\mu, \sigma^2) \Rightarrow f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$

multivariate: $X \sim N(\mu, \Sigma) \Rightarrow f_X(x) = \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}} \exp\left(\frac{(x-\mu)^T \Sigma^{-1}(x-\mu)}{2}\right)$

μ → $[x_1 x_2 \cdots x_n]$ cov

Σ → cov → tell the shape of elliptical contour of gaussian surface (horizontal slice)

eigenval & eigenvector → Principal axis of ellipse

LDA ① Discriminative vs Generative Paradigms & Bayes Decision rule And GDA's MLE derivation:

in discriminative we use all data pt but learn separate class data

But

in generative we all distribution of data for a given class & that are almost independent to classes of each other.

Binary classification:

$y^{(i)} \in \{0, 1\}$

Given

$P(X|Y=0; \phi) \sim N(\mu_0, \Sigma_0)$

$(X|Y=1; \phi) \sim N(\mu_1, \Sigma_1)$

$Y \sim \text{Bernoulli}(\phi)$

in the LDA model we assume all the centroids of different classes from their centers are of same shape (Σ same)

$\Sigma_0 = \Sigma_1 = \Sigma$

$X = [x_1, x_2, \cdots, x_n]$ features

multivariate — no need for $\perp$

$P(X|Y=0) = \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu_0)^T \Sigma^{-1}(x-\mu_0)\right)$

$P(X|Y=1) = \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu_1)^T \Sigma^{-1}(x-\mu_1)\right)$

$P(Y=1) = \phi$

$P(Y=y^{(i)}) = \phi^{y^{(i)}}(1-\phi)^{(1-y^{(i)})}$

$P(X|Y=y^{(i)}) = \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu_{y^{(i)}})^T \Sigma^{-1}(x-\mu_{y^{(i)}})\right)$

Likelihood $L(\phi)$ $\quad \phi \rightarrow$ set of parameters $\rightarrow \phi, \Sigma, \mu_0, \mu_1$

$L(\phi) = P(\vec{Y}|\vec{X};\phi) =$

$= \prod_{i=1} P(y^{(i)}|x^{(i)};\phi)$

$= \prod_{i=1} \left(\frac{P(x^{(i)}|y^{(i)}) P(y^{(i)})}{P(x^{(i)})}\right)$

same for all classes $P(X) = \sum_{i=1}^{K} P(X|Y=y^{(i)}) P(Y=y^{(i)})$

$\log L(\phi) = l(\phi) = \sum \left(\log P(x^{(i)}|y^{(i)}) + \log(P_{y^{(i)}}) + P(x^{(i)})\right)$

$$\ell(\phi) = \sum_{i=1}^{m} \Big(\log\big(p(x^{(i)} | y^{(i)}; \phi)\big) + \log\big(p(y^{(i)})\big) - \log\big(p(x^{(i)})\big) \Big)$$

$$\ell(\phi) = \sum_{i=1}^{m} \Big(y^{(i)} \log(\phi) + (1 - y^{(i)}) \log(1-\phi) \Big) - \sum \log p(x^{(i)})$$

$$\frac{m_0}{2}\log 2\pi - \frac{m}{2}\log|\Sigma| - \frac{1}{2}(x^{(i)} - \mu)^T \Sigma^{-1} (x^{(i)} - \mu)$$

3 IM

$$\ell(\phi) = \sum_{i=1}^{m} \big(y^{(i)} \log\phi + (1 - y^{(i)}) \log(1-\phi) \big) + C$$

$$\frac{\partial \ell(\phi)}{\partial \phi} = \sum_{i=1}^{m} \Big(\frac{y^{(i)}}{\phi} - \frac{(1-y^{(i)})}{(1-\phi)} \Big) = 0 \quad \phi - m\phi - \Sigma\phi = 0$$

$$\phi - \phi\phi - \phi + y\phi$$

$$\phi = \frac{\sum_{i=1}^{m} y^{(i)}}{m}$$

$$(m - x^{(i)})^T \Sigma^{-1} (x^{(i)} - \mu)$$

$$\ell(\mu_0) = -\frac{1}{2} \sum_{i: y^{(i)}=0} \Big((x^{(i)} - \mu_0)^T \Sigma^{-1} (x^{(i)} - \mu_0) \Big)$$

$$\frac{\partial \ell(\mu_0)}{\partial \mu_0} = -\frac{1}{2} \sum_{i: y^{(i)}=0} \Big(-2\Sigma^{-1} x + 2\Sigma^{-1} \mu_0 \Big) = 0$$

$$(x - \mu_0)^T A (x - \mu_0)$$

$$(x^T - \mu_0^T)(Ax - A\mu_0) = x^T A x - x^T A \mu_0 - \mu_0^T A x + \mu_0^T A \mu_0$$

$$-2(A^T x)^T \qquad 2\mu_0^T A \frac{\partial \mu_0}{\partial \mu_0}$$

$$\sum_{i: y^{(i)}=0} (x - \mu_0) = 0$$

$$\mu_0 = \sum_{i=1}^{m} 1\{y^{(i)} = 0\} \qquad \mu_1 = \sum_{i=1}^{m} 1\{y^{(i)} = 1\}$$

$$m_0 + m_1 = m$$

$$\mu_0 = \frac{\sum_{i: y^{(i)}=0} x^{(i)}}{m_0} \qquad \mu_1 = \frac{\sum_{i: y^{(i)}=1} x^{(i)}}{m_1}$$

$$x^{(i)} \in \mathbb{R}^{m_1} \quad \mu_0 \in \mathbb{R}^{m_1}$$

$$\ell(\Sigma) = C - \frac{m}{2}\log|\Sigma| - \frac{1}{2}\sum_{i=1}^{m} (x^{(i)} - \mu_{y^{(i)}})^T \Sigma^{-1} (x^{(i)} - \mu_{y^{(i)}})$$

$$\operatorname{tr}(A) = \operatorname{tr}(A^T)$$

$$\ell(\Sigma) = C - \frac{m}{2}\log|\Sigma| - \frac{1}{2}\operatorname{tr}(\Sigma^{-1} Z^T Z)$$

$$\frac{\partial \ell(\Sigma)}{\partial \Sigma} = \frac{m}{2}\Sigma^{-1} - \frac{1}{2}(Z^T Z)^T \Sigma^{-1} = 0$$

$$\Sigma = \frac{Z^T Z}{m}$$

$$\Sigma = \frac{\sum_{i=1}^{m} (x^{(i)} - \mu_{y^{(i)}})(x^{(i)} - \mu_{y^{(i)}})^T}{m}$$

Date / Page /

④ $\Sigma = \frac{1}{m} \sum_{i=1}^{m} (x^{(i)} - \mu_{y^{(i)}})(x^{(i)} - \mu_{y^{(i)}})^T$

What we are doing is finding centres for different classes & then from those centers we make our gaussians with same covariance i.e. a shape & when they meet give us straight line.

$\Sigma_k = \Sigma$ for all this make linear line fit possible



↓ LDA fit method

to predict

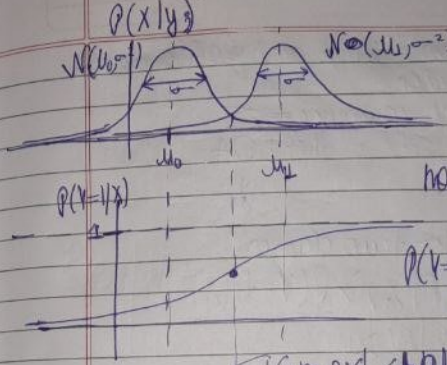
$\underset{k}{\text{argmax}}\ P(y|x;\theta) = \underset{k}{\text{argmax}}\ P(x|y)\,P(y)$

$k \in \{0,1\}$

$\underset{k}{\text{argmax}} \left(P(x|y=y^{(k)})\, P(y=y^{(k)}) \right)$

now suppose if we have two classes at centres μ_0 & μ_1 lets see its geometrical shifts $P(y|x)$ curve

Date / Page /



Sigmoid shape

now $P(Y=1|X) = \frac{P(X|Y=1) \times P(y=1)}{P(X)}$

$P(Y=1) = P(Y=0)$

$\Rightarrow \frac{P(X|Y=1)}{P(X|Y=1) + P(X|Y=0)}$

eg LDA

$(X|Y=0) \sim N(\mu_0, \Sigma)$
$(X|Y=1) \sim N(\mu_1, \Sigma)$
$(y=1) \sim \text{Bernoulli}(\theta)$

} imply ⟹ logistic regression

Sigmoid $P(Y=1|X)$

• strong assumption	• weak assumption
• more restricted by data	• general type of model not dependent on data type
• more accurate	• well for large data
• but on data type change our accuracy drops more	
• well for small data	

Overall

$(X|Y=0) \sim$ exp class
$(X|Y=1) \sim$ same exp class
$(y=1) \sim \text{Bernoulli}(\theta)$

} imply ⟹ Sigmoid f^n $P(Y=1|X)$

* if we have K classes then x each follows the gaussian distribution
and have same covariance $\Rightarrow \Sigma$
then LDA

if we compare for two classes

$$P(Y=f|X) > P(Y=t|X)$$

$$\frac{P(Y=f|X)}{P(Y=t|X)} > 1$$

$$\log\left(\frac{P(Y=f|X)}{P(Y=t|X)}\right) > 0$$

Do, basically need $\log\left(\frac{P(Y=f|X)}{P(Y=t|X)}\right)$

$P(Y=k|X) \sim N(\mu_k, \Sigma)$

$$\frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}} e^{-\frac{1}{2}(x-\mu_k)^T\Sigma^{-1}(x-\mu_k)}$$

$$-\left((x-\mu_f)^T\Sigma^{-1}(x-\mu_f) - (x-\mu_t)^T\Sigma^{-1}(x-\mu_t)\right)$$

$(x^T-\mu_f^T)(x-\mu_f) = x^Tx - x^T\mu_f - \mu_f^Tx + \mu_f^T\mu_f$ ①

$= x^Tx - 2x^T\mu_f + \mu_f^T\mu_f$

Do, $-\left(x^T\Sigma^{-1}x - x^T\Sigma^{-1}\mu_f - \mu_f^T\Sigma^{-1}x + \mu_f^T\Sigma^{-1}\mu_f\right)$ ① Σ^{-1} ②

$-\left(x^Tx - 2x^T\Sigma^{-1}\mu_f + \mu_f^T\Sigma^{-1}\mu_f - x^Tx + 2x^T\Sigma^{-1}\mu_t - \mu_t^T\Sigma^{-1}\mu_t\right)$

$+ 2x^T\Sigma^{-1}(\mu_f - \mu_t) + \mu_t^T\Sigma^{-1}\mu_t - \mu_f^T\Sigma^{-1}\mu_f$

$(\mu_t^T + \mu_f^T)\Sigma^{-1}(\mu_t - \mu_f)$

$\mu_t^T\Sigma^{-1}\mu_t - \mu_f^T\Sigma^{-1}\mu_f$

$-\mu_t^T\Sigma^{-1}\mu_f + \mu_f^T\Sigma^{-1}\mu_t$

$$\log\left(\frac{P(Y=f|X)}{P(Y=t|X)}\right) =$$

$= 2x^T\Sigma^{-1}(\mu_f - \mu_t) + (\mu_f + \mu_t)^T\Sigma^{-1}(\mu_f - \mu_t)$

linear in term of x in LDA $\Sigma(x) = \Sigma$

→ if we want to see the score $\delta_k(x)$

$\delta_k(x) = \log(P(Y=k|X;\theta))$

$= \log(P(X|Y=k)) + \log(P(Y=k)) - \log(P(X=x))$ (constant)

$= -\frac{n}{2}\log 2\pi - \frac{1}{2}\log|\Sigma| - \log(P(X=x)) - \frac{1}{2}(x-\mu_k)^T\Sigma^{-1}(x-\mu_k) + \log P(Y=k)$

$= \log\phi_k - \frac{1}{2}(x-\mu_k)^T\Sigma^{-1}(x-\mu_k) + C$

$= \log\phi_k - \frac{1}{2}(x^T\Sigma^{-1}x - 2x^T\Sigma^{-1}\mu_k + \mu_k^T\Sigma^{-1}\mu_k)$ ← same for all

$\boxed{\text{argmax}_k\ \delta_k(x)}$

$\delta_k(x) = \log\phi_k - \frac{1}{2}(\mu_k^T\Sigma^{-1}\mu_k - 2x^T\Sigma^{-1}\mu_k) + C$

$= \log\phi_k - \frac{1}{2}(\mu_k^T - 2x^T)\Sigma^{-1}\mu_k + C$

$\boxed{\delta_k(x) = x^T\Sigma^{-1}\mu_k - \frac{1}{2}\mu_k^T\Sigma^{-1}\mu_k + \log\phi_k + C}$

$\log(P(Y=k|X))$

$$\log\left(\frac{P(Y=a|x)}{P(Y=b|x)}\right) = \log(P(Y=a|x)) - \log(P(Y=b|x))$$
$$= \delta_a(x) - \delta_b(x)$$

$$= \mu_a^T\Sigma^{-1}x - \mu_b^T\Sigma^{-1}x - \frac{1}{2}\mu_a^T\Sigma^{-1}\mu_a + \frac{1}{2}\mu_b^T\Sigma^{-1}\mu_b + \log\frac{\phi_a}{\phi_b}$$

$$= x^T\Sigma^{-1}(\mu_a-\mu_b) - \frac{1}{2}\left((\mu_a+\mu_b)^T\Sigma^{-1}(\mu_a-\mu_b)\right) + \log\frac{\phi_a}{\phi_b}$$

$=0$ for decision Boundary

If $P(x) \sim N(\mu, \sigma^2)$ equal priors

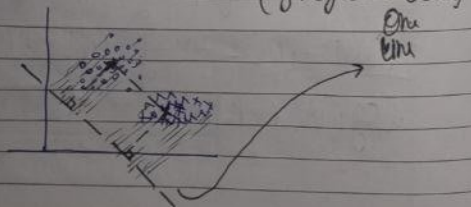
$$\frac{x}{\sigma^2}(\mu_1-\mu_2) = \frac{1}{2}\frac{(\mu_1+\mu_2)}{\sigma^2}(\mu_1-\mu_2)$$

$$x = \frac{\mu_1+\mu_2}{2}$$

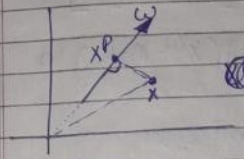


Fishers Method

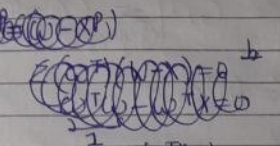
What we want to do is that maximize the between class distance relative to in class variance.



If we have two classes ① & ②



$x^p = zw$
$\|w\| = 1$



$(x-x^p) \perp w$

$$w^T(x-zw) = 0$$
$$w^Tx - zw^Tw = 0$$
$$z = w^Tx \rightarrow \text{scale}$$

If we have m_1 centre $\frac{\sum_{i=1}^{m} \mathbb{1}(y^{(i)}=1)x^{(i)}}{\sum_{i=1}^{m}\mathbb{1}(y^{(i)}=1)}$

m_2 centroid $= \frac{\sum \mathbb{1}(y^{(i)}=2)x^{(i)}}{\sum \mathbb{1}(y^{(i)}=2)}$

If we project centroid on w line

$m_1^p = w^Tm_1$ so each point $x^{(i)} = w^Tx^{(i)}$
$m_2^p = w^Tm_2$ average $w^T\frac{\sum(\ )}{m_1} = w^Tm_1$

If we compare dis just

$m_1^p - m_2^p = w^T(m_1-m_2)$

if scale the $w^T(m_1-m_2)$ no difference

scatter of variance in class

$$s_1^2 = \sum_{i:y^{(i)}=1}(w^Tm_1 - w^Tx^{(i)})^2 = w^T\left(\sum_{i=y^{(i)}=1}(x_i-m_1)^2\right)w$$

$$S_1^2 = \omega^T \left[\sum_{i:y(i)=1} (x^{(i)} - m_1)(x^{(i)} - m_1)^T \right] \omega$$

$$S_2^2 = \omega^T \left[\sum_{i:y(i)=2} (x^{(i)} - m_2)(x^{(i)} - m_2)^T \right] \omega$$

$S_1^2 = \omega^T C_1 \omega$, $S_2^2 = \omega^T C_2 \omega$

Squared dist
Inclastar

$$\frac{(m_1 - m_2)^2}{S_1^2 + S_2^2} = \frac{\omega^T (\mu_1 - \mu_2)(\mu_1 - \mu_2)^T \omega}{\omega^T (C_1 + C_2) \omega}$$

$$= \frac{\omega^T S_B \omega}{\omega^T S_W \omega}$$

$$\text{argmax}_{\omega} \frac{\omega^T S_B \omega}{\omega^T S_W \omega}$$

$$f(\omega) = \frac{\omega^T A \omega}{\omega^T B \omega}$$

$$f'(\omega) = 0 \Rightarrow 2B(\omega)(\omega^T A \omega) - 2D\omega(\omega^T A \omega) = 0$$

$$A\omega = \frac{\omega^T A \omega}{\omega^T B \omega} B\omega = f(\omega) B\omega$$

$$A\omega = \lambda B\omega$$

$$S_B = (\mu_1 - \mu_2)(\mu_1 - \mu_2)^T = dd^T$$

$$dd^T\omega = \lambda B\omega$$

$$d = \lambda' B\omega$$

$$B^{-1}d = \lambda'\omega$$

$$\boxed{\omega \propto S_W^{-1}(\mu_1 - \mu_2)}$$

QDA: $\Sigma(k)$ for each class k

$$\delta_k(x) = \log \phi_k - \frac{1}{2} x^T \Sigma_k^{-1} x + x^T \Sigma_k^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k$$

$$\log(P(y=k|x)) = \delta_k(x) = -\frac{1}{2} x^T \Sigma_k^{-1} x - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + x^T \Sigma_k^{-1} \mu_k + \log \phi_k - \frac{1}{2} \log|\Sigma_k|$$

$\delta_1(x) = \delta_2(x)$ boundary

$$-\frac{1}{2} x^T \Sigma_1^{-1} x - \frac{1}{2} \mu_k^T \Sigma_1^{-1} \mu_k + x^T \Sigma_1^{-1} \mu_k + \log \phi_1 - \frac{1}{2} \log|\Sigma_1| =$$

$$-\frac{1}{2} x^T \Sigma_2^{-1} x - \frac{1}{2} \mu_2^T \Sigma_2^{-1} \mu_2 + x^T \Sigma^{-1} \mu_2 + \log \phi_2 - \frac{1}{2} \log|\Sigma_2|$$

quadratic terms

Naive Bayes

if we have a dictonary of 10,000 words

Bernolli each word present or absent

dict = $[x_1\ x_2 \ldots\ldots\ldots\ldots\ldots\ldots\ x_{10,000}]^T$

Classify, $y \in \{0, 1\}$ ⇒ spam, ham (not spam)

Presence of x_j given $y=0$

- $\phi_{j|y=0}$
- $\phi_{j|y=1}$

Presence of x_j given $y=1$

- $\phi_{y=1} = \phi_y$

absence of x_j given $y=0$

- $1 - \phi_{j|y=0}$
- $1 - \phi_{j|y=1}$

Absence of x_j given $y=1$

- $1 - \phi_y$

$P(Y=y^{(i)}) = \phi_y^{y^{(i)}} (1-\phi_y)^{(1-y^{(i)})}$

$P(x_1, x_2, \dots, x_{10000} | y) = P(x_1|y)\, P(x_2|x_1, y) \cdots P(x_{10000}|x_1 \cdots x_{9999}, y)$

$x_1, x_2, \dots, x_{10000}$ are independent

$P(x_1, x_2, \dots, x_{10000}|y) = \prod_{i=1}^{10000} P(x_i|y)$

$L(\phi) = P(\vec{y}|\vec{x};\phi) = \prod P(y^{(i)}|x^{(i)};\phi)$

$\Rightarrow P(y^{(i)}|x^{(i)};\phi) = \frac{P(x^{(i)}|y^{(i)};\phi)\, P(y^{(i)})}{P(x^{(i)})}$ → same

$L(\phi) = \prod_{i=1}^{m} P(x^{(i)}|y^{(i)};\phi)\, P(y^{(i)})$

$l(\phi) = \sum_{i=1}^{m} \left(\sum_{j=1}^{10000} \log(\phi_{j|y^{(i)}}) \right) + y^{(i)} \log \phi_y + (1-y^{(i)}) \log(1-\phi_y)$

$\phi \Rightarrow \phi_{j|0}, \phi_{j|1}, \phi_y$

$l(\phi_y) = \sum_{i=1}^{m} \left(y^{(i)} \log \phi_y + (1-y^{(i)}) \log(1-\phi_y) \right) + C$

$\frac{\partial l(\phi_y)}{\partial \phi_y} = \sum_{i=1}^{m} \left(\frac{y^{(i)}}{\phi_y} - \frac{(1-y^{(i)})}{1-\phi_y} \right) = 0$

$y - y\phi - \phi + \phi y$

$\sum_{i=1}^{m} y^{(i)} - m\phi = 0 \Rightarrow \boxed{\phi_y = \frac{\sum_{i=1}^{m} y^{(i)}}{m}}$

$P(x_j^{(i)}|y^{(i)}) =$

$\left((\phi_{j|1})^{x_j^{(i)}} (1-\phi_{j|1})^{1-x_j^{(i)}} \right)^{y^{(i)}} \left((\phi_{j|0})^{x_j^{(i)}} (1-\phi_{j|0})^{1-x_j^{(i)}} \right)^{1-y^{(i)}}$

$\left(\sum_{j=1}^{10,000} \left(x_j^{(i)} \log(\phi_{j|1}) + (1-x_j^{(i)}) \log(1-\phi_{j|1}) \right) y^{(i)} \right)$

$+ \sum_{j=1}^{10000} \left(x_j^{(i)} \log(\phi_{j|0}) + (1-x_j^{(i)}) \log(1-\phi_{j|0}) \right)(1-y^{(i)}) \Bigg)$

$l(\phi_{j|0}) = \sum_{i=1}^{m} \sum_{j=1}^{10,000} (1-y^{(i)}) \left(x_j^{(i)} \log(\phi_{j|0}) + (1-x_j^{(i)}) \log(1-\phi_{j|0}) \right)$

$\frac{\partial l(\phi_{j|0})}{\partial \phi_{j|0}} = \sum_{i=1}^{m} \left[(1-y^{(i)}) \left(\frac{x_j^{(i)}}{\phi_{j|0}} - \frac{(1-x_j^{(i)})}{1-\phi_{j|0}} \right) \right] = 0$

$\frac{x_j^{(i)} - \phi_{j|0}}{\phi_{j|0}(1-\phi_{j|0})}$

$\sum_{i=1}^{m} (1-y^{(i)})(x_j^{(i)} - \phi_{j|0}) = 0$

$\boxed{\phi_{j|0} = \frac{\sum_{i=1}^{m} (1-y^{(i)}) x_j^{(i)}}{\sum_{i=1}^{m} (1-y^{(i)})}} \qquad \boxed{\phi_{j|1} = \frac{\sum_{i=1}^{m} y^{(i)} x_j^{(i)}}{\sum_{i=1}^{m} y^{(i)}}}$

$$\phi_{j|y=1} = \frac{\sum_{i=1}^{m} y^{(i)} x_j^{(i)}}{\sum_{i=1}^{m} y^{(i)}} = \frac{\sum_{i=1}^{m} \mathbb{1}(y^{(i)}=1)\, x_j^{(i)}}{\sum_{i=1}^{m} \mathbb{1}(y^{(i)}=1)}$$

$$\phi_{j|y=1} = \frac{\sum_{i:y=1} x_j^{(i)}}{\sum_{i:y=1} 1} \quad | \quad \phi_{j|y=0} = \frac{\sum_{i:y=0} x_j^{(i)}}{\sum_{i:y=0} 1}$$

GDA: $(X|Y=y^{(i)}) \sim \mathcal{N}(\mu_{y^{(i)}}, \Sigma)$

in $y^{(i)} \in \{0,1\}$ Binary classification $\quad \Sigma \rightarrow$ covariance matrix

$\mu_0, \mu_1, \Sigma, \phi$

$\phi = P(y=1)$

$m = m_0 + m_1$

$$\phi = \frac{\sum_{i=1}^{m} y^{(i)}}{m}; \quad m_0 = \sum_{i=1}^{m} \mathbb{1}(y^{(i)}=0); \quad m_1 = \sum_{i=1}^{m} \mathbb{1}(y^{(i)}=1)$$

$$\mu_0 = \frac{\sum_{i:y^{(i)}=0} x^{(i)}}{m_0}; \quad \mu_1 = \frac{\sum_{i:y^{(i)}=1} x^{(i)}}{m_1}; \quad \Sigma = \frac{1}{m}\sum_{i=1}^{m} (x^{(i)}-\mu_{y^{(i)}})(x^{(i)}-\mu_{y^{(i)}})^T$$

$\partial|\Sigma| = |\Sigma|\,\mathrm{tr}(\Sigma^{-1}\partial\Sigma)$

$\Sigma\Sigma^{-1} = I$

$k \in \{0,1\}$ $\quad \operatorname{argmax}_k \left(P(x|y=y^{(i)})\, P(y=y^{(i)})\right)$

form of linear eqn

If we have K classes then can compare their score $\delta_k(x)$

$$\delta_k(x) = \log(P(Y=k|x)) = \mu_k^T \Sigma^{-1} x - \frac{1}{2}\mu_k^T \Sigma^{-1} \mu_k + \log \phi_k + C$$

$$\delta_k(x) = x^T \Sigma^{-1} \mu_k - \frac{1}{2}\mu_k^T \Sigma^{-1} \mu_k + \log \phi_k + C$$

find $\operatorname{argmax}_k \delta_k(x)$

$$\Sigma = \frac{1}{m}\sum_{i=1}^{m} \left[(x^{(i)}-\mu_{y^{(i)}})(x^{(i)}-\mu_{y^{(i)}})^T\right]$$

$$\mu_k = \frac{\sum_{i=1}^{m} \mathbb{1}(y^{(i)}=k)\, x^{(i)}}{\sum_{i=1}^{m} \mathbb{1}(y^{(i)}=k)}; \quad \phi_k = \frac{\sum_{i=1}^{m} y^{(i)} \mathbb{1}(y^{(i)}=k)}{m}$$

GDA: $\Sigma^{(k)}$ each have quadratic term or decision surface

$$\delta_k(x) = -\frac{1}{2}\log|\Sigma_k| + \log \phi_k - \frac{1}{2} x^T \Sigma_k^{-1} x - \frac{1}{2}\mu_k^T \Sigma_k^{-1} \mu_k + x^T \Sigma_k^{-1} \mu_k$$

Fischer scale: $\frac{(m_1-m_2)^2}{S_1^2+S_2^2}$ for projected data

$S_1^2 \rightarrow$ scatter matrices for class 1

$$\frac{(m_1-m_2)^2}{S_1^2+S_2^2} = \frac{\omega^T S_B \omega}{\omega^T S_W \omega}$$

$$S_1^2 = \omega^T \left[\sum_{i:y^{(i)}=1} (x^{(i)}-m_1)(x^{(i)}-m_1)^T\right]\omega$$

$$S_2^2 = \omega^T \left[\sum_{i:y^{(i)}=2} (x^{(i)}-m_2)(x^{(i)}-m_2)^T\right]\omega$$

$$S_1^2 + S_2^2 = \omega^T (C_1+C_2)\omega = \omega^T S_N \omega$$

$$S_B = (\mu_1-\mu_2)(\mu_1-\mu_2)^T$$

$$\omega \propto S_\omega^{-1}(\mu_1-\mu_2)$$

Naive Bayes

Binary: 1 or 0

$$\phi_{j|0} = \frac{\sum_{i:y=0} x_j^{(i)}}{\sum_{i:y=0} 1}, \quad \phi_{j|1} = \frac{\sum_{i:y=1} x_j^{(i)}}{\sum_{i:y=1} 1}$$

$$\phi_{y=1} = \frac{\sum_{i:y^{(i)}=1} y^{(i)}}{m}$$

QDA

$$\delta_k(x) = -\frac{1}{2}\log|\Sigma_k| + \log\phi_k - \frac{1}{2}x^T\Sigma_k^{-1}x - \frac{1}{2}\mu_k^T\Sigma_k^{-1}\mu_k + x^T\Sigma_k^{-1}\mu_k$$

$$\Sigma_k = \frac{1}{n_k}\sum_{i:y^{(i)}=k}(x^{(i)}-\mu_k)(x^{(i)}-\mu_k)^T$$

$$n_k = \sum_{i=1}^{m} 1(y^{(i)}=k)$$

Quadratic surface

* In symmetric dxd matrix we have $d+(d-1)+(d-2)\cdots+1 = \frac{d(d+1)}{2}$ uniquetems

$\Sigma \rightarrow$ dxd matrix

LDA → $\mu_1 \cdots \mu_k \rightarrow k$
$\Sigma \rightarrow d(d+1)/2$
$\phi_1, \phi_2 \cdots, \phi_k \rightarrow k$
} $2k + \frac{d(d+1)}{2}$ parametrs

QDA → $\mu_1 \cdots \mu_k \rightarrow k$
$\Sigma_1, \Sigma_2 \cdots, \Sigma_k \rightarrow kd(d+1)/2$
$\phi_1, \phi_2 \cdots, \phi_k \rightarrow k$
} $2k + \frac{kd(d+1)}{2}$ parametrs

* Lets see different types of Naive Bayes

Naive Bayes // we have K classes

$$\arg\max_k P(Y=k|X)$$

$$P(Y=k|X) = \frac{P(X|Y=k)\,P(Y=k)}{P(X)}$$

$$P(X) = P(X|Y=1)P(Y=1) + P(X|Y=2)P(Y=2) \cdots + P(X|Y=K)P(Y=K)$$

↳ same for all classes

$$\arg\max_k \left(\phi_k P(X|Y=k)\right)$$

Assume in naive Bayes we assume the conditional independence rather then marginal independence

✓ $P(x_1, x_2, \cdots, x_d | Y=k) = \prod_{i=1}^{d} P(X_i|Y=k)$

✗ not $P(x_1, x_2, \cdots, x_d) \neq \prod_{i=1}^{d} P(x_i)$

① Gaussian Bayes $(x_j | Y=k) \sim N(\mu_{jk}, \sigma_{jk}^2)$

$$P(X^{(i)}|Y=k) = \prod_{i=1}^{d} P(X_i|Y=k)$$

$$= \prod_{i=1}^{d} \frac{1}{\sqrt{2\pi}\sigma_{ik}} \exp\left(\frac{-(x_i-\mu_{ik})^2}{2\sigma_{ik}^2}\right)$$

② Bernoulli

$x_j^{(i)} \in \{0,1\}$

$$P(X^{(i)}|Y=k) = \prod_{j=1}^{d} \phi_{kj}^{x_j}(1-\phi_{kj})^{(1-x_j)}$$

$\phi_{kj} = P(X_j=1|Y=k)$

③ Multinomial Naive Bayes

x_j is count of appearance rather than 0 or 1

$\phi_{jk} = P(j\text{ word} \mid Y=k)$

$P(x \mid Y=k) \propto \prod_{j=1}^{d} \phi_{jk}^{x_j}$

Bernoulli NaiveBayes

$x_j^{(i)} \in \{0,1\}$ $\quad \{(x^{(i)}, y^{(i)})\}_{i=1}^{m}$

$P(y=k) = \pi_k$ $\quad y \in k$ classes

$P(x \mid y=k) = \prod_{j=1}^{d} \phi_{kj}^{x_j} (1-\phi_{kj})^{(1-x_j)}$

$\phi_{kj} = P(x_j=1 \mid Y=k)$

$L(\pi, \phi) = \prod_{i=1}^{m} P(y^{(i)}) P(x^{(i)} \mid y^{(i)})$

$L = \prod_{i=1}^{m} \left(P(y^{(i)}) \left(\prod_{j=1}^{d} \phi_{y^{(i)}j}^{x_j^{(i)}} (1-\phi_{y^{(i)}j})^{1-x_j^{(i)}} \right) \right)$

$l = \sum_{i=1}^{m} \log \pi_{y^{(i)}} + \sum_{i=1}^{m} \sum_{j=1}^{d} \left(x_j^{(i)} \log \phi_{y^{(i)}j} + (1-x_j^{(i)}) \log (1-\phi_{y^{(i)}j}) \right)$

$\sum \pi_k = 1$

$l_\pi = \sum_{i=1}^{m} \log \pi_{y^{(i)}}$

$n_k = \sum_{i=1}^{m} 1(y^{(i)}=k)$

$\pi_k = \frac{n_k}{m}$

$l(\phi_{kj}) = \sum_{i:y^{(i)}=k} \left[x_j^{(i)} \log \phi_{kj} + (1-x_j^{(i)}) \log (1-\phi_{kj}) \right]$

$\phi_{kj} = \frac{\sum_{i:y=k} x_j^{(i)}}{n_k}$

④ Laplace Smoothing

if $P(x_j=1 \mid Y=k) = 0$

from train data (event not occur)

if test $x_j=1$

$\Rightarrow P(X \mid Y=k) = \prod P(x_j \mid Y=k) = 0$

$P(Y=k \mid x) = 0$

Unseen shows 0 probability

Laplace (Add-α) Smoothing :

$\pi_k = \frac{n_k + \alpha}{m + \alpha C}$ $\quad C$ = no. of possible category (no. of class)

$\phi_{kj} = \frac{n_{kj} + \alpha}{n_k + C\alpha}$

$n_{kj} = \sum_{i:y^{(i)}=k} (x_j^{(i)})$

$n_k = \sum 1(y^{(i)}=k)$

Date
Page

$$P(X|Y=k) = \prod_{j=1}^{d} P(x_j|Y=k)$$

$$\hat{\pi}_k = \frac{n_k}{m} \; ; \; n_k = \sum_{i:} \mathbb{1}(y^{(i)}=k)$$

$$\hat{\theta}_{kj} = \frac{1}{n_k} \sum_{i:y=k} x_j^{(i)}$$

$$\hat{\theta}_{kj}^{\text{laplace}} = \frac{n_{kj}+\alpha}{n_k+C\alpha}$$

$$\underset{k}{\operatorname{argmax}} \left(\log \pi_k + \sum_j \log P(x_j|Y=k) \right)$$

$$n_{kj} = \sum_{i:y(i)=k} x_j^{(i)}$$

$$P(x_j^{(i)}|Y=k) = \theta_{kj}^{x_j^{(i)}} (1-\theta_{kj})^{(1-x_j^{(i)})}$$

$$\theta_{kj} = P(X_j=1|Y=k)$$